IEEE P3335 Proposal for Architecture and Interfaces for Time Card

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Contents

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Abstract

This draft specifies a generic architecture and interface framework for TimeCard systems that provide time-of-day, phase, and frequency services to directly attached and networked systems. It defines a unified-timescale model, timing and control interface requirements, performance-reporting requirements, environmental declarations, conformance profiles, traceability-supporting evidence, and test guidance. The framework accommodates discrete cards, embedded functions, and external timing units while providing common information needed for integration and comparison.

Keywords: clock synchronization, frequency stability, holdover, IEEE P3335, precision time, TimeCard, timing interface, traceability

Introduction

The TimeCard concept has been developed through contributions to IEEE P3335 and related work in the Open Compute Project Time Appliances Project. This draft organizes that material as an implementation-neutral architecture, a set of externally observable interface requirements, and a common basis for declaring performance.

The text distinguishes requirements needed for base conformance from conditional requirements that apply only when an interface or profile is claimed. Informative annexes provide metric background, example test procedures, a bibliography, and a conformance-statement proforma.

1. Overview

1.1 Scope

This standard defines the generic architecture and interfaces of a TimeCard system, which constitutes a traceable source of time-of-day to heterogeneous systems that distribute or use that time. It also defines figures of merit that characterize the relevant performance of a TimeCard.

A TimeCard provides time-of-day to directly attached systems and to networked distributed systems. Such systems include a host containing the TimeCard and systems synchronized through protocols such as the Precision Time Protocol (PTP) or Network Time Protocol (NTP). P3335 requires the reference relationships, measurement points, uncertainty information, and supporting evidence needed to evaluate the TimeCard contribution to a system-level traceability analysis. Conformance to P3335 does not, by itself, establish end-to-end traceability for an integrated system.

This standard defines the basic functional building blocks of a TimeCard and the interfaces among those blocks to support modular implementation. The principal functions include a time source, a local timing function, and a time processor.

This standard also defines physical and logical interfaces between a TimeCard and other systems. Physical interfaces convey time-related input and output signals. Logical interfaces support host operating-system integration, including access to a PTP hardware clock (PHC). The logical model accommodates form factors such as Peripheral Component Interconnect Express (PCIe), embedded hardware functions, and external timing units.

Conforming implementations provide performance figures obtained and reported according to this standard so that TimeCard implementations can be evaluated on a common basis.

1.2 Purpose

The purpose of this standard is to provide:

- Interoperability between TimeCard implementations and the systems and operating systems that use them as sources of time-of-day.
- Modular implementation of TimeCard functions to support different application needs and implementation technologies.
- Unambiguous comparison of TimeCard implementations using common performance metrics, measurement points, and operating conditions.

1.3 Need

Distributed computing, telecommunications, industrial control, finance, and scientific measurement can depend on precise and traceable time. Existing implementations often use product-specific signal definitions, control models, and performance claims. These differences increase integration effort and make independent comparison difficult.

P3335 addresses that need by defining externally observable TimeCard functions, conditional interface profiles, a unified-timescale model, a transport-neutral control information model, and a common performance declaration framework.

1.4 Applicability and implementation freedom

This standard applies to a TimeCard realized as a discrete add-in card, embedded hardware or gateware, a system-on-chip function, or an external timing unit. Conformance is based on observable behavior, implemented interfaces, and supplier declarations rather than physical form factor.

This standard does not prescribe an internal oscillator technology, disciplining algorithm, ensemble algorithm, physical partitioning, or product construction unless that choice affects a claimed interface or externally observable behavior. It does not replace product-safety, electromagnetic-compatibility, environmental, spectrum, export, or other regulatory obligations.

1.5 Relationship to other standards

P3335 uses established standards where they define a protocol, signal, or metric needed by a claimed TimeCard function. Clause 2 lists the documents that are indispensable to implementing those functions. Annex C lists related documents used for background or deployment guidance.

The principal areas of coordination are:

- IEEE Std 1588-2019 [4] and IEEE Std 802.1AS-2025 [5] for packet-based time synchronization.
- IEEE Std 1139-2022 [2], IEEE Std 1193-2022 [3], ITU-T Recommendation G.810 (08/1996) [10], and ITU-T Recommendation G.8260 (11/2022) [11] for time and frequency metrology.
- The PCI Express Base Specification, Revision 5.0, Version 1.0 [13], for PCIe and Precision Time Measurement behavior.
- IRIG Standard 200-16 [8] for claimed IRIG time-code interfaces.
- Management protocol specifications for control interfaces that claim those protocols.

1.6 Document structure

Clause or annex	Content
Clause 1	Scope, purpose, need, applicability, and document organization.
Clause 2	Normative references.
Clause 3	Terms, definitions, acronyms, and abbreviations.
Clause 4	Conformance model, profiles, statements, and evidence.
Clause 5	TimeCard architecture and unified-timescale behavior.
Clause 6	Performance declaration and characterization requirements.
Clause 7	Receive and providing timing interfaces.
Clause 8	Control interfaces and the baseline control information model.
Clause 9	Environmental, mechanical, electrical, and lifecycle declarations.
Clause 10	Informative application and deployment guidance.
Annex A	Informative metric background.
Annex B	Informative example test procedures.
Annex C	Bibliography.
Annex D	Informative conformance-statement proforma.

2. Normative References

The following referenced documents are indispensable for the application of this document. Only the dated edition, amendment, or corrigendum identified in each entry applies. A later edition does not apply unless it is incorporated by an amendment or revision of this standard.

Informative references and background material are listed in Annex C.

2.1 Standards and specifications

- [1] **DMTF DSP0236**, *Intelligent Platform Management Interface Specification, Second Generation*, Version 2.0, 20 February 2015.
- [2] **IEEE Std 1139-2022**, *IEEE Standard Definitions of Physical Quantities for Fundamental Frequency and Time Metrology–Random Instabilities*, 2022.

- [3] **IEEE Std 1193-2022**, *IEEE Guide for Measurement of Environmental Sensitivities of Frequency Standards*, 2022.
- [4] **IEEE Std 1588-2019**, *IEEE Standard for a Precision Clock Synchronization Protocol for Networked Measurement and Control Systems*, 2019.
- [5] **IEEE Std 802.1AS-2025**, *IEEE Standard for Local and Metropolitan Area Networks–Timing and Synchronization for Time-Sensitive Applications*, 2025.
- [6] **IETF RFC 3411**, *An Architecture for Describing Simple Network Management Protocol (SNMP) Management Frameworks*, December 2002.
- [7] **IETF RFC 5905**, *Network Time Protocol Version 4: Protocol and Algorithms Specification*, June 2010.
- [8] **IRIG Standard 200-16**, *IRIG Serial Time Code Formats*, August 2016.
- [9] **ITU-T Recommendation G.703 (04/2016)**, *Physical/electrical characteristics of hierarchical digital interfaces*, including Amendment 1 (05/2021).
- [10] **ITU-T Recommendation G.810 (08/1996)**, *Definitions and terminology for synchronization networks*, including Corrigendum 1 (11/2001).
- [11] **ITU-T Recommendation G.8260 (11/2022)**, *Definitions and terminology for synchronization in packet networks*, including Amendment 1 (01/2024).
- [12] **MIPI Alliance Specification for I3C Basic, Version 1.2**, 16 December 2024.
- [13] **PCI-SIG PCI Express Base Specification, Revision 5.0, Version 1.0**, 28 May 2019.
- [14] **System Management Bus (SMBus) Specification, Version 3.2**, 12 January 2022.

2.2 Reference use

Normative references are cited in the body of this standard at the point where their content is required to implement an interface, measurement method, or reported characteristic. A referenced document that is used only for examples, background, or deployment guidance is listed in Annex C instead of this clause.

3. Definitions, Acronyms, and Abbreviations

3.1 Definitions

For the purposes of this document, the following terms and definitions apply.

- **accuracy:** Qualitative closeness of agreement between a measured quantity value and a reference quantity value. A numerical accuracy claim is expressed in this standard as time error, frequency offset, uncertainty, or another defined metric.
- **command queue:** Ordered control-interface mechanism through which a producer submits commands or requests and a consumer reports completion, status, or response data according to a defined mapping.
- **conformance profile:** Named set of mandatory requirements that applies when the profile is claimed by an implementation.
- **conformance statement:** Supplier documentation identifying the P3335 clauses, profiles, optional features, interfaces, limits, and evidence for which an implementation claims conformance.
- **control interface:** Logical or physical interface used to configure, monitor, update, or manage a TimeCard.
- **control plane:** Set of logical or physical paths and functions used to configure, monitor, update, or manage a TimeCard.
- **data plane:** Logical or physical path or set of paths that conveys time, phase, frequency, or event-timestamp information.
- **declared:** Stated in the conformance statement or in supplier documentation referenced by that statement.
- **disciplining:** Steering a local timing function using observations of one or more reference signals.
- **ensemble operation:** Combination of selected reference observations to produce a synthesized reference used to discipline or validate the unified timescale.

- **evidence:** Records, observations, calculations, test results, calibration information, or controlled documentation used to support evaluation of a conformance claim.
- **granularity:** Smallest step representable by an encoding or interface. Granularity does not imply an equal resolution, precision, or accuracy.
- **holdover:** Operating state in which a TimeCard maintains its timescale after loss or rejection of the external reference used for synchronization, using its local timing function and available historical information.
- **host clock:** Clock maintained or exposed by a host system and identified by its clock domain, epoch, timescale, and adjustment behavior.
- **host system:** Computing, telecommunications, measurement, or control platform that integrates or consumes the services of a TimeCard.
- **host-time correlation:** Bounded observation that associates a TimeCard timestamp with host-clock readings taken around the same TimeCard measurement operation.
- **instance identifier:** Mapping-scoped identifier used to select one TimeCard when one or more TimeCard instances are accessible through the same host or control environment. The identifier has a declared stability scope.
- **lock time:** Elapsed time from a declared starting condition until a TimeCard enters the declared locked state.
- **measurement point:** Physical connector, logical boundary, register access point, packet event, or other defined location to which a measured or declared value applies.
- **measurement uncertainty:** Non-negative parameter characterizing the dispersion of quantity values attributed to a measurand.
- **optional feature:** Feature, interface, protocol, environmental profile, or security function not required for base conformance that becomes subject to its applicable mandatory requirements when claimed.
- **phase alignment:** Specified phase relationship between corresponding timing markers on two interfaces. Alignment error is the measured difference from that relationship.
- **precision:** Closeness of agreement among repeated indications under stated conditions.
- **providing interface:** Outbound physical or logical interface through which a TimeCard distributes time, phase, frequency, or timestamp information.
- **receive interface:** Inbound physical or logical interface and its defined measurement point through which a TimeCard acquires observations of an external reference signal or data stream.
- **reference signal:** External physical signal or data stream whose defined timing marker or encoded time value can be used to initialize, discipline, validate, or monitor the unified timescale.
- **resolution:** Smallest change in a measured quantity that causes a perceptible change in the corresponding indication.
- **serial interface:** Control interface that transfers commands and data sequentially over a physical or virtual serial channel using framing and command syntax defined by its mapping.
- **time error:** Difference between the time indicated by the TimeCard at a specified measurement point and the time of the declared reference timescale at the corresponding instant.
- **TimeCard:** Modular timing subsystem that maintains a unified timescale and provides time, phase, frequency, or timestamp services through one or more interfaces.
- **time jitter:** Short-term variation of a specified timing event from its ideal position after removal of the trend or deterministic components identified by the declared measurement method.
- **timescale:** Ordered system of time values with a defined reference, epoch, and rules for forming time intervals.
- **time-control ownership:** Arbitration state that grants one authorized client exclusive permission to perform operations that set or discipline the unified timescale or its local timing function.
- **traceability:** Property of a measurement result by which the result can be related to a stated reference through a documented, unbroken chain of calibrations, each contributing to measurement uncertainty. In this standard, a TimeCard supplies declarations and evidence that support system-level traceability analysis; the term does not assert that a TimeCard alone establishes end-to-end traceability.
- **unavailable:** Status indicating that a supported object or measurement does not currently have a valid value.
- **unified timescale:** Single timescale maintained by a TimeCard instance from which all of that in-

stance's providing interfaces derive their time, phase, frequency, or timestamp information.

- **unspecified:** Status or behavior that P3335 or an applicable mapping intentionally does not constrain, leaving the selection or definition to an implementation, profile, or later specification. For an object field, it may also indicate that the mapping permits a value but the implementation has not assigned one. `unspecified` is distinct from `unknown`, `unavailable`, and `unsupported`.
- **unsupported:** Status indicating that an object, operation, or feature is not implemented.
- **valid:** Status indicating that a value satisfies the freshness, state, and integrity conditions defined by its interface mapping.

3.2 Acronyms and Abbreviations

- **1PPS:** one pulse per second
- **ADEV:** Allan deviation
- **ABI:** application binary interface
- **AM:** amplitude modulation
- **API:** application programming interface
- **ASIC:** application-specific integrated circuit
- **BAR:** base address register
- **BMC:** baseboard management controller
- **CFM:** cubic feet per minute
- **CSAC:** chip-scale atomic clock
- **DCLS:** direct current level shift (IRIG time-code modulation)
- **DDS:** direct digital synthesizer
- **DMTF:** Distributed Management Task Force
- **EMC:** electromagnetic compatibility
- **EMI:** electromagnetic interference
- **ESD:** electrostatic discharge
- **FPGA:** field-programmable gate array
- **GLONASS:** Global Navigation Satellite System of the Russian Federation
- **GNSS:** global navigation satellite system
- **GPIO:** general-purpose input/output
- **GPS:** Global Positioning System
- **gRPC:** Google Remote Procedure Call
- **I2C:** Inter-Integrated Circuit
- **I3C:** Improved Inter-Integrated Circuit
- **ID:** identifier
- **IEEE:** Institute of Electrical and Electronics Engineers
- **IEEE SA:** IEEE Standards Association
- **IETF:** Internet Engineering Task Force
- **IP:** Internet Protocol
- **IPMI:** Intelligent Platform Management Interface
- **IRIG:** Inter-Range Instrumentation Group
- **ITU-T:** International Telecommunication Union Telecommunication Standardization Sector
- **LFM:** linear feet per minute
- **LOS:** loss of signal
- **mdeg C:** millidegrees Celsius
- **MMIO:** memory-mapped input/output
- **MIPI:** MIPI Alliance; originally Mobile Industry Processor Interface, but MIPI is no longer expanded as an acronym
- **MTBF:** mean time between failures
- **MTIE:** maximum time interval error
- **NC-SI:** Network Controller Sideband Interface
- **NavIC:** Navigation with Indian Constellation
- **NMI:** national metrology institute

- **ns:** nanosecond
- **NTP:** Network Time Protocol
- **OCP:** Open Compute Project
- **OCP TAP:** Open Compute Project Time Appliances Project
- **OCXO:** oven-controlled crystal oscillator
- **PCI:** Peripheral Component Interconnect
- **PCIe:** Peripheral Component Interconnect Express
- **PCI-SIG:** Peripheral Component Interconnect Special Interest Group
- **PHC:** PTP hardware clock
- **PLL:** phase-locked loop
- **ps:** picosecond
- **PPS:** pulse per second
- **PTM:** Precision Time Measurement
- **PTP:** Precision Time Protocol
- **QZSS:** Quasi-Zenith Satellite System
- **REST:** representational state transfer
- **RF:** radio frequency
- **RFC:** Request for Comments
- **RMS:** root mean square
- **SMA:** SubMiniature version A coaxial connector
- **SMBus:** System Management Bus
- **SNMP:** Simple Network Management Protocol
- **TAI:** International Atomic Time
- **TAP:** Time Appliances Project
- **TCXO:** temperature-compensated crystal oscillator
- **TDEV:** time deviation
- **TIC:** time interval counter
- **TIE:** time interval error
- **TLS:** Transport Layer Security
- **TNC:** Threaded Neill-Concelman coaxial connector
- **ToD:** time of day
- **Type-N:** Type-N coaxial connector
- **µs:** microsecond
- **USB:** Universal Serial Bus
- **UTC:** Coordinated Universal Time
- **UTF-8:** Unicode Transformation Format, 8-bit
- **WR:** White Rabbit
- **WWVB:** United States long-wave time-signal station WWVB

4. Conformance

This clause defines the conditions under which an implementation may claim conformance to IEEE P3335. A conforming implementation **shall** satisfy the mandatory requirements in this clause and the mandatory requirements in Clauses 5 through 9 that apply to the implemented features, interfaces, and operating conditions.

Informative clauses, informative notes, and informative annexes provide explanatory material only. They do not create conformance requirements unless a normative clause explicitly references a requirement stated elsewhere.

4.1 Requirement Terms

The following terms are used as defined by IEEE SA rules for standards text:

- **shall** indicates a mandatory requirement.

- **should** indicates a recommended practice among several permissible possibilities.
- **may** indicates a course of action permitted within the limits of this standard.
- **can** indicates capability, possibility, or a statement of fact.

The term **must** is not used as a requirement term in this standard. Where a physical or mathematical inevitability needs to be described, the text should be written as a statement of fact using **is**, **are**, or **can**.

The term **unspecified** is not a requirement term. It identifies a value, behavior, or design choice that P3335 has considered but intentionally does not constrain. The choice is left to the implementation, an applicable profile, or a later specification, subject to any applicable documentation requirements.

4.2 Conformance Model

IEEE P3335 uses interspersed requirements with a conformance summary. Mandatory and optional requirements are stated in the technical clauses where the relevant behavior is specified. Clause 4 summarizes how those requirements are applied.

A TimeCard implementation may be realized as a discrete add-in card, an embedded module, an FPGA or ASIC function, a system-on-chip block, or an external timing unit. The form factor does not determine conformance. Conformance is determined by the externally observable interfaces, behavior, documentation, and performance declarations of the implementation.

A claim of conformance **shall** identify at least the following:

- The implementation form factor and host interface mapping.
- The timing receive and providing interfaces implemented.
- The control interfaces implemented.
- The optional feature sets claimed.
- The declared operating environment and environmental limits.
- The performance metrics, measurement points, measurement methods, and limits declared by the supplier.

4.3 Base Conformance Requirements

A conforming TimeCard implementation **shall** meet all of the following base requirements:

- The implementation **shall** provide at least one timing output providing interface that distributes time, phase, frequency, or a combination thereof to a host system or downstream system.
- All implemented timing output providing interfaces **shall** derive from the single unified timescale specified in Clause 5.
- The implementation **shall** provide at least one accessible control interface as specified in Clause 8.
- The implementation **shall** document all implemented timing interfaces, control interfaces, physical connectors, protocol mappings, and optional features.
- The implementation **shall** document the performance metrics required by Clause 6 for each applicable measurement point.
- The implementation **shall** document the environmental limits and reliability information required by Clause 9 for the declared deployment environment.
- The implementation **shall** define the conditions under which each declared performance value is valid, including warm-up time, lock time, reference source, temperature range, and measurement bandwidth where applicable.
- The implementation **shall** not claim support for an optional feature unless all mandatory requirements associated with that feature are satisfied.

An implementation may conform to IEEE P3335 without implementing every optional receive interface, providing interface, host interface mapping, management protocol, environmental profile, or security hardening feature described in this standard.

4.4 Conformance Profiles

This standard defines the following conformance profiles. An implementation claiming a profile **shall** satisfy the corresponding additional requirements in the following table.

Profile	Applicability	Additional requirements
Base TimeCard	All conforming implementations	Satisfy the base requirements in 4.3 and all applicable mandatory requirements in Clauses 5 through 9.
Physical Timing Output	Implementations with externally accessible physical timing outputs	Provide a 1PPS output that satisfies 7.3.2 and document the alignment of all other physical timing outputs to that 1PPS output.
PCIe Host Mapping	Implementations claiming PCIe host integration	Satisfy the PCIe host interface profile requirements in 8.9.
Managed TimeCard	Implementations claiming remote or fleet management	Satisfy the managed security profile in 8.11.2.
Secure Infrastructure TimeCard	Implementations marketed or designated for secure infrastructure deployments	Satisfy the secure infrastructure profile in 8.11.3.

An embedded or logical-only TimeCard implementation may claim Base TimeCard conformance without a physical 1PPS connector if it does not expose externally accessible physical timing outputs. Such an implementation **shall** still provide at least one timing output providing interface and **shall** document the equivalent measurement point for timing-performance declarations.

4.5 Optional Feature Claims

Optional features are conditional. If an implementation claims support for an optional feature, the implementation **shall** satisfy the mandatory requirements that apply to that feature.

Optional feature claims include, but are not limited to:

- Receive interfaces such as GNSS, PTP, NTP, White Rabbit, WiWi, WWVB, IRIG, PPS, or frequency-reference inputs.
- Providing interfaces such as PPS, frequency outputs, Time of Day outputs, PTP, PCIe PTM, or other host-bus time-transfer mechanisms.
- Host interface mappings such as PCIe, USB, serial, embedded memory-mapped interfaces, or implementation-specific mappings.
- Control protocols such as SMBus, I2C, I3C, IPMI, NC-SI, REST, gRPC, SNMP, Redfish, serial, or USB.
- Security functions such as authenticated management, signed firmware update, secure boot, encrypted telemetry, key storage, sanitization, or tamper response.
- Environmental profiles such as data-center, telecommunications, industrial, laboratory, or field deployment.
- Ensemble reference processing, holdover classes, or other enhanced timing functions.

Each optional feature claim **shall** identify the clause or subclause that defines the claimed behavior and **shall** identify any deviations or implementation-specific limits allowed by that clause.

4.6 Performance Conformance Strategy

Base conformance to this standard requires the bounded and reproducible performance reporting specified in Clause 6. It does not require a universal numeric minimum time-accuracy, frequency-stability, phase-noise, jitter, or holdover class.

An implementation may claim a vendor-defined or application-defined performance class. If such a class is claimed, the conformance statement **shall** identify the class, the metric limits, the measurement points, and the conditions under which the class applies.

4.7 Conformance Statement

The supplier of a conforming implementation **shall** provide a conformance statement. The conformance statement **shall** be both publicly available and supplied with the product documentation and **shall** include, at a minimum:

- Product name, hardware revision, firmware revision, and relevant configuration profile.
- Claimed IEEE P3335 conformance scope.
- Supported timing receive interfaces and providing interfaces.
- Supported host interface mapping and control interface classes.
- Host driver, service, API, and ABI revisions required for each claimed host mapping.
- Instance-identifier stability, host lifecycle behavior, and host-time correlation capabilities for each claimed host mapping.
- Claimed conformance profiles from 4.4.
- Supported optional feature sets.
- Declared performance metrics and measurement points.
- Declared environmental operating and storage limits.
- References to datasheets, register maps, interface descriptions, calibration information, and test reports needed to evaluate the claim.

The conformance statement should be written so that an independent test laboratory or system integrator can determine which requirements apply without relying on unpublished implementation details.

4.8 Test Evidence

Conformance can be evaluated by functional testing, performance testing, documentation review, or a combination thereof.

Functional testing verifies that the implemented interfaces, state transitions, control operations, and fault responses behave as specified. Performance testing verifies that declared timing, frequency, holdover, and environmental metrics meet the supplier's stated limits using the measurement methods declared for the implementation.

When test evidence is used to support a conformance claim, the evidence **shall** identify:

- The device under test and configuration under test.
- The applicable IEEE P3335 clauses and optional feature claims.
- Test equipment, calibration status, and traceability path.
- Test environment, reference source, cabling, and measurement points.
- Pass/fail criteria and measured results.
- Any deviations, waivers, or limitations of the test method.

Annex B provides informative example test procedures that can be used as a starting point for such evidence. Its procedures do not add, change, or replace the normative conformance criteria.

4.9 Conformance Statement Proforma

Annex D provides an informative proforma for organizing the conformance statement required by 4.7. Use of the proforma is optional. The requirements in this clause and the applicable technical clauses remain the controlling source for conformance.

5. Architecture (Normative)

This clause defines the architectural requirements for a TimeCard. The architecture is specified in terms of externally observable functions, interfaces, behavior, and documentation. A TimeCard may be implemented as a discrete card, an embedded hardware block, an FPGA or ASIC function, an external module, or another implementation form that satisfies the requirements of this standard.

The architecture separates timing data-plane functions from management and control functions. Timing data-plane functions receive, generate, maintain, timestamp, and provide time, phase, and frequency. Management and control functions configure, monitor, secure, and report the state of those timing functions. Figure 1 shows the principal external relationships.

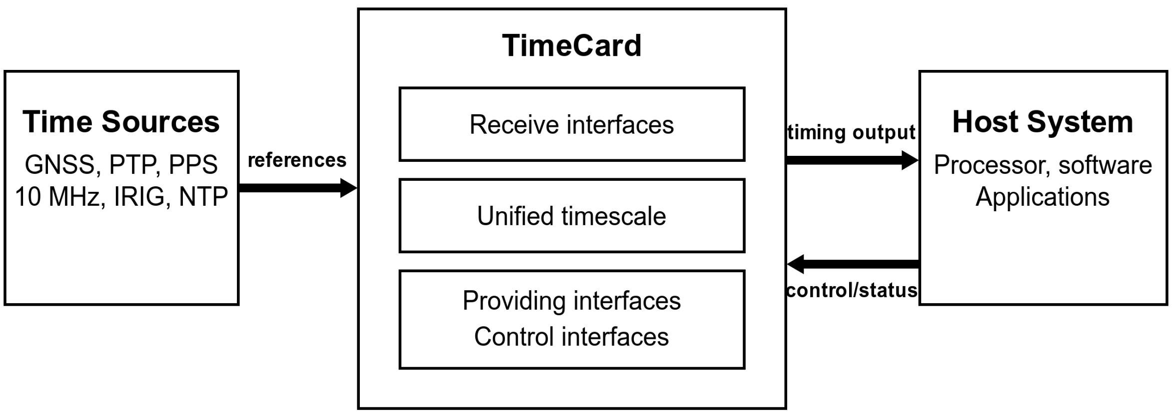

Figure 1: TimeCard context within a host system

5.1 Architectural Overview

A TimeCard is a timing subsystem that provides time, phase, frequency, or a combination thereof to a host system or downstream system. The host system may be a server, telecommunications platform, industrial controller, scientific instrument, embedded system, or other system that requires a timing service with documented support for system-level traceability analysis.

A conforming TimeCard **shall** include the following externally observable capabilities:

- A local timing function capable of maintaining a unified timescale.
- At least one timing output providing interface.
- At least one control interface.
- Published performance, interface, environmental, and conformance information sufficient for an implementer or operator to evaluate the claimed behavior.

A TimeCard may include one or more timing receive interfaces, one or more time-transfer interfaces, one or more host interface mappings, and one or more reference-selection or ensemble algorithms.

5.2 Implementation Boundary

The TimeCard implementation boundary is the set of interfaces through which the TimeCard exchanges timing signals, timestamp values, control information, status, telemetry, firmware, or power with the host system or external equipment.

An implementation **shall** document its implementation boundary. The documentation **shall** identify:

- Host interface mappings.
- Timing receive interfaces, if any.
- Timing providing interfaces.
- Control interfaces.
- Power, thermal, and mechanical dependencies.
- Measurement points used for declared performance values.

Internal partitioning is implementation-specific unless a claimed optional feature or interface mapping makes that partitioning externally observable.

5.3 Core Functional Blocks

A TimeCard architecture is described by the following logical blocks. An implementation may combine or subdivide these blocks, provided that the externally observable behavior remains conformant.

5.3.1 Local Timing Function

The local timing function maintains the TimeCard's time, phase, and frequency state. It may be based on a quartz oscillator, OCXO, TCXO, CSAC, rubidium oscillator, atomic reference, DDS, host-provided clock, or another timing source.

The implementation **shall** document the oscillator or timing-source class used for each declared performance profile. The implementation **shall** document whether the local timing function is steerable, free-running, disciplined by external references, or operated as a primary reference.

5.3.2 Time Generator

The time generator produces the TimeCard's unified timescale. It may include counters, phase accumulators, timestamp interpolation, leap-second or timescale conversion logic, frequency steering, phase steering, or other timing logic.

The time generator **shall** support timestamp representations that permit unambiguous duration computation over the declared operating interval. If an implementation exposes a timescale that is not TAI, UTC, or the IEEE 1588-2019 [4] PTP timescale, the implementation **shall** document the mapping to the declared reference timescale.

The implementation **shall** document the supported timestamp range, epoch, granularity, resolution, and behavior at discontinuities such as leap seconds, leap smearing, holdover entry, reference failover, or manual time steps.

5.3.3 Timestamping Facility

If an implementation timestamps events, packets, register accesses, signal edges, or host-bus transactions, the timestamping facility **shall** derive those timestamps from the unified timescale. The implementation **shall** document the timestamping measurement point, timestamp latency or correction model, resolution, granularity, and known fixed offsets.

Hardware timestamping should be used for receive and providing interfaces where the supported protocol or physical interface permits it.

5.3.4 Receive Interfaces

Receive interfaces acquire external references that may be used to steer, discipline, initialize, monitor, or validate the unified timescale. Receive interfaces are specified in Clause 7.

An implementation may operate without a receive interface if it is designed for free-running, primary-reference, laboratory, or isolated holdover operation. If no receive interface is implemented, the conformance statement **shall** state that no external timing reference is supported.

5.3.5 Providing Interfaces

Providing interfaces distribute time, phase, frequency, or timestamp information from the TimeCard to a host system or downstream system. Providing interfaces are specified in Clause 7.

At least one providing interface **shall** be implemented. All providing interfaces implemented by a single TimeCard **shall** derive from the same unified timescale.

5.3.6 Control Interfaces

Control interfaces provide configuration, monitoring, alarms, telemetry, firmware management, and security operations. Control interfaces are specified in Clause 8.

At least one control interface **shall** be implemented. A control interface may be local, host-integrated, out-of-band, remote, or a combination thereof.

5.4 Unified Timescale

A TimeCard **shall** generate one unified timescale for each conforming TimeCard instance. All providing interfaces of that instance **shall** publish, encode, or derive from that unified timescale. Figure 2 illustrates the reference-to-egress flow.

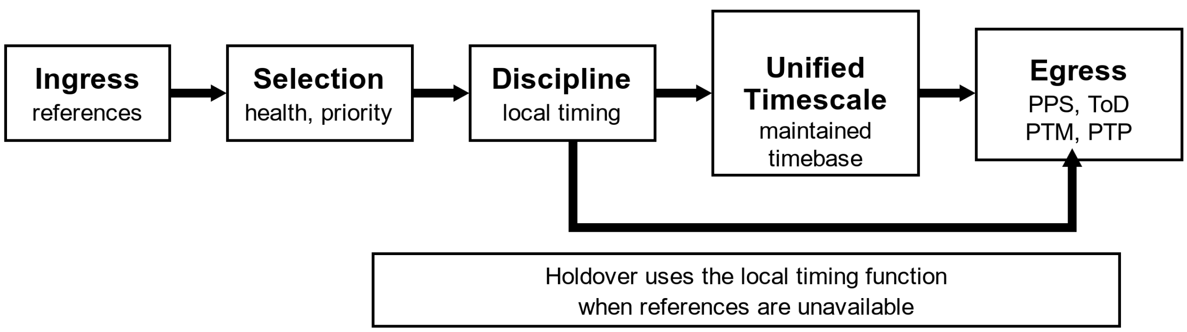

Figure 2: Unified timescale flow

The implementation **shall** document:

- The reference timescale used internally or exposed externally.
- The relationship between the unified timescale and UTC, TAI, the IEEE 1588 PTP timescale, or another declared timescale.
- The alignment tolerance among all providing interfaces that claim phase or time alignment.
- The behavior during startup, warm-up, lock acquisition, reference failover, holdover entry, holdover exit, and manual time adjustment.

Analog or periodic providing interfaces that claim continuous phase **shall** not introduce an undocumented phase step during normal locked operation. If a phase step, holdover event, failover transient, or other discontinuity can occur, its detection and reporting mechanism **shall** be documented.

5.5 Reference Selection and Ensemble Operation

If more than one receive reference is implemented, the TimeCard **shall** provide a documented reference-selection policy. The policy **shall** identify priority, validity, quality, failure detection, fallback, recovery, and operator override behavior.

If an implementation claims ensemble reference operation, it **shall** document:

- Supported input reference types.
- Weighting, voting, or selection method at a level sufficient for operational evaluation.
- Health and alarm criteria.
- Behavior when one or more ensemble sources degrade or fail.
- Metrics exposed through the control interface.

Ensemble operation **shall** produce one selected or synthesized reference for disciplining the unified timescale. It **shall** not create multiple independent conforming timescales within the same TimeCard instance unless those instances are separately identified and separately documented.

5.6 Host and Time-Transfer Interfaces

A host interface mapping defines how a host system discovers, configures, reads, writes, timestamps, or receives time from a TimeCard. PCIe, USB, serial, memory-mapped embedded interfaces, network interfaces, and implementation-specific mappings may be used.

Each host interface mapping **shall** document:

- Physical and logical interface standards used.
- Discovery and enumeration behavior.
- Per-instance selection, instance-identifier derivation, and identifier stability across restart, removal, reinsertion, and host reboot.
- Required driver-visible registers, messages, commands, or API elements.
- Endianness, alignment, units, and timestamp formats.
- Interrupts, polling requirements, and error reporting.
- Concurrency, time-control ownership, and behavior when more than one host client can issue timing-affecting operations.
- Behavior during host low-power states, sleep, hibernation, wake, orderly or surprise removal, hot plug, driver or service restart, and outstanding-operation cancellation.
- Latency, asymmetry, and correction information required for time-transfer evaluation.
- Host-clock identity, epoch, timescale, discontinuity behavior, and freshness information used for host-time correlation.

A host interface mapping that can expose more than one TimeCard **shall** provide unambiguous per-instance selection. An enumeration index **shall** not be the sole persistent selector when a stable supplier-assigned or implementation-derived identity is available.

A mapping that implements host-time correlation **shall** preserve the semantics and validity rules in 8.10.4. A mapping that permits more than one client to set or discipline time **shall** implement the arbitration requirements in 8.5.

If PCIe PTM is implemented, the implementation **shall** document the PTM capability defined by the PCI Express Base Specification, Revision 5.0, Version 1.0 [13], the measurement point represented by PTM timestamps, and any correction terms needed to relate PTM time to other providing interfaces.

Annex E provides informative examples of applying these requirements to Windows, macOS, and Linux host software.

5.7 Performance and Measurement Architecture

Performance requirements in this standard are expressed as reporting and measurement requirements unless a technical clause explicitly states a numeric limit. This allows TimeCards to serve different use cases while making supplier claims comparable.

An implementation **shall** identify the measurement point for each declared metric. Metrics **shall** be reported with enough context to reproduce or evaluate the claim, including reference source, measurement bandwidth, observation interval, environmental conditions, warm-up conditions, and lock state.

The following performance categories **shall** be addressed when applicable to the implemented interfaces:

- Time accuracy relative to a declared reference.
- Time stability, including MTIE and TDEV where applicable.
- Frequency stability, including ADEV where applicable.
- Phase noise for periodic frequency outputs where applicable.
- Time jitter for pulse outputs where applicable.
- Holdover error versus elapsed time.
- Lock acquisition, recovery, and failover behavior.
- Environmental sensitivity relevant to the declared operating profile.

Clause 6 defines the required reporting details for these categories. Annex A provides informative metric background.

5.8 Power, Mechanical, and Environmental Architecture

Physical TimeCard implementations **shall** document the power, mechanical, thermal, connector, and environmental assumptions needed for safe operation and declared timing performance.

Embedded or virtualized TimeCard implementations **shall** document the host resources, clock dependencies, thermal assumptions, and integration constraints that affect declared timing performance.

Detailed environmental requirements are specified in Clause 9.

5.9 Optional Features and Extensions

An implementation may include optional features or vendor-specific extensions. An extension **shall** use the documented extension mechanism for the applicable interface and **shall** not change the syntax, semantics, required state transitions, or error behavior of a standardized interface or profile claimed by the implementation.

Vendor-specific extensions **shall** be identified by a documented namespace, identifier range, capability, or version that distinguishes standardized P3335 behavior from implementation-specific behavior. Reserved values, register fields, message types, and identifiers **shall** not be assigned to extension behavior.

5.10 Documentation Requirements

Manufacturers **shall** provide documentation sufficient to evaluate and integrate a conforming TimeCard. The documentation **shall** include:

- Conformance statement required by Clause 4.
- Interface descriptions for all receive, providing, host, and control interfaces.
- Register maps, information models, schemas, command descriptions, or API descriptions needed to use the implemented control interfaces.
- PLL or disciplining-loop type and bandwidth, or a statement that the implementation does not use such a loop.
- Warm-up time to first usable output and warm-up time to declared full performance.
- Lock acquisition time, failover behavior, holdover entry and exit behavior, and recovery behavior.
- Declared performance metrics and measurement points required by Clause 6.
- Environmental and reliability information required by Clause 9.
- Reference-chain, calibration, measurement-point, and uncertainty information sufficient to support system-level traceability analysis, or an explicit statement identifying which information is unavailable.
- All optional features claimed and all conditional requirements exercised by those claims.

Documentation should be publicly available for commercial products and available to integrators for non-commercial, embedded, or custom implementations.

5.11 Informative Time-Flow Narrative

In a typical TimeCard, one or more receive interfaces acquire external timing references such as GNSS, PTP, PPS, frequency references, or other sources. Reference-selection logic evaluates those inputs and selects or synthesizes a reference. A disciplining function steers the local timing function. The time generator maintains the unified timescale. Providing interfaces distribute that unified timescale as PPS, frequency, Time of Day, packet timestamps, host-bus time, or other representations. Control interfaces report status, telemetry, alarms, and configuration state.

When all usable external references are lost, the TimeCard normally enters holdover or free-running operation. During holdover, the unified timescale continues according to the local timing function and any holdover model implemented by the TimeCard. The TimeCard reports the state transition and exposes the information needed to evaluate the declared holdover behavior.

6. Performance Specifications (Normative)

This clause defines the performance reporting and characterization requirements for TimeCard implementations. Base conformance does not define a single universal numeric performance class for all use cases. Instead, it requires performance claims to be stated using common metrics, defined measurement points, traceable methods, and declared operating conditions.

6.1 Performance Declaration Model

Suppliers **shall** publish performance declarations for each conforming TimeCard implementation. Each declared value **shall** identify:

- The metric being reported.
- The measurement point.
- The operating mode, such as locked, holdover, free-running, warm-up, failover, or recovery.
- The reference source or reference timescale.
- The applicable temperature range, voltage range, airflow or cooling condition, and other environmental conditions.
- The minimum warm-up time and lock time required before the declaration applies.
- The observation interval, averaging time, bandwidth, sample count, or statistical confidence basis used.
- The measurement equipment class and traceability path.

Each required performance declaration **shall** include at least one bounded value or limit. Typical values may be provided in addition to, but not instead of, the bounded value.

6.2 Performance Classes

This standard uses a reporting-only base performance model. A TimeCard conforms to the base performance requirements when it reports the applicable metrics in this clause using the declared measurement methods and operating conditions.

Named numeric performance classes are not defined by this standard. A supplier, procurement profile, or application profile may define additional numeric classes. If an implementation claims such a class, the class **shall** identify all applicable metric limits, measurement points, operating states, and environmental conditions.

6.3 General Measurement Requirements

Measurement results used to support conformance claims **shall** be metrologically traceable to a recognized national metrology institute or another declared reference through a documented calibration chain. Measurement methods **shall** conform to the applicable definitions in IEEE Std 1139-2022 [2], IEEE Std 1193-2022

[3], ITU-T Recommendation G.810 (08/1996) [10], or ITU-T Recommendation G.8260 (11/2022) [11] when those metrics are used.

The measurement uncertainty, instrument noise floor, measurement bandwidth, averaging configuration, sample count, data treatment, and environmental conditions **shall** be documented with the reported result. The measurement point **shall** be identified in enough detail that another laboratory can reproduce the setup.

When a measured result is compared with a declared limit, the test report **shall** state the decision rule and treatment of measurement uncertainty. A pass/fail conclusion **shall** not silently treat an uncertainty interval that crosses the limit as an unqualified pass.

If the term **jitter** is used, the implementation documentation **shall** define the exact jitter metric, measurement bandwidth, sample population, and statistical calculation. Time jitter, phase noise, MTIE, TDEV, and ADEV **shall** not be used interchangeably.

6.4 Time Accuracy

Time accuracy declarations **shall** state the error of a TimeCard output or timestamp relative to a declared reference timescale or source.

For each time accuracy declaration, the supplier **shall** document:

- Source of standard time, such as GNSS system time, UTC(k), TAI, PTP grandmaster time, or another declared reference.
- Measurement point, such as 1PPS output, Time of Day output, PTP egress timestamp, PTM timestamp, or register-read timestamp.
- Maximum time error or another explicitly defined bounded statistic.
- Statistical basis, such as maximum observed value, percentile, RMS value, or confidence interval.
- Static error and dynamic error components when they are separately known.
- Valid temperature range and environmental profile.
- Minimum lock time before the declaration applies.
- Relationship between the declared source and UTC or TAI, if the declared source is not itself UTC or TAI.

If an implementation reports IEEE 1588-2019 [4] `clockAccuracy` values, the values **shall** correspond to the measured or declared accuracy range for the applicable operating mode.

6.5 Time Stability

Time stability declarations **shall** characterize variation of the TimeCard time output or timestamp stream over a defined observation interval.

Suppliers **shall** report MTIE for time outputs or timestamp streams for which bounded time-error behavior is claimed. The MTIE declaration **shall** include observation intervals, measurement point, lock state, environmental conditions, and applicable limit or measured result.

Suppliers should report TDEV for time outputs or timestamp streams where stochastic time deviation is operationally relevant. TDEV declarations **shall** include averaging intervals and the method used to compute the result.

6.6 Frequency Stability and Phase Noise

Frequency stability declarations **shall** characterize the frequency output or local timing function over one or more averaging intervals.

Suppliers **shall** report frequency accuracy and temperature stability for the local timing function or for each frequency output for which a frequency-performance claim is made.

Suppliers should report ADEV for frequency outputs or oscillator functions where long-term or short-term frequency stability is relevant. ADEV declarations **shall** include the averaging intervals and measurement point.

For periodic outputs such as 10 MHz, suppliers should report phase noise as a function of offset frequency. Phase-noise declarations **shall** identify carrier frequency, offset-frequency range, measurement bandwidth, instrument configuration, and measurement point.

6.7 Pulse and Event Timing

For pulse outputs such as 1PPS, suppliers **shall** document:

- On-time edge definition.
- Output polarity.
- Pulse width range.
- Rise and fall time measurement method.
- RMS time jitter or another explicitly defined short-term timing variation metric.
- Peak-to-peak or bounded time variation when claimed.
- Alignment to other providing interfaces of the same TimeCard.

For event-capture or timestamping inputs, suppliers **shall** document the event measurement point, timestamp resolution, granularity, latency, fixed corrections, and known uncertainty contributors.

6.8 Holdover Performance

Holdover declarations **shall** characterize accumulated time error after loss of all external synchronization references being used.

For each holdover declaration, the supplier **shall** document:

- Holdover entry condition and triggering event.
- Reference source and lock duration before holdover begins.
- Minimum warm-up and stabilization conditions before holdover begins.
- Measurement point.
- Maximum time error or MTIE versus elapsed holdover time.
- Temperature profile, sensor location, rate of change, dwell conditions, and input-voltage conditions during holdover.
- Behavior when the reference is restored, including qualification, reacquisition, phase and frequency transients, continuity, and the condition for returning to the locked state.

MTIE as defined by ITU-T Recommendation G.810 (08/1996) [10] or ITU-T Recommendation G.8260 (11/2022) [11] **shall** be used when reporting bounded holdover time error. Other holdover indicators, such as aging rate or estimated drift, may be reported in addition to MTIE.

6.9 Dynamic Operation

Suppliers **shall** characterize timing behavior during operational transitions that can affect output phase, frequency, or timestamp values. The characterization **shall** include, where applicable:

- Cold-start and warm-start lock acquisition time.
- Transition from locked operation to holdover.
- Transition from holdover to locked operation.
- Reference failover between two valid sources.
- Manual time step or frequency steering command.
- Firmware update or restart behavior if timing service is interrupted or degraded.
- Host entry into and recovery from a low-power, sleep, or hibernation state.
- Orderly removal, surprise removal, hot-plug disconnection, and reinsertion of a host-connected TimeCard.

- Driver, host service, or time-provider unload, reload, failure, and restart.
- A host-clock discontinuity that invalidates a host-time correlation or a sample awaiting delivery.

For each transition, the supplier **shall** document whether the unified timescale remains continuous and whether a discontinuity can occur in phase or in a time derivative of phase. The documentation **shall** bound applicable phase steps, frequency steps, frequency-slope changes, kinks or corners, and recovery transients, and **shall** identify how each condition is reported through the control interface.

6.10 Environmental and host-system effects

Performance declarations **shall** identify the environmental and host-system conditions under which the declared values apply. The following effects **shall** be characterized when applicable to the declared deployment profile:

- Temperature and thermal-gradient sensitivity.
- Input-voltage variation and supply noise sensitivity.
- Vibration sensitivity.
- Airflow and host thermal loading.
- Host-bus activity, electromagnetic interference, or crosstalk that can affect timing outputs.
- Host scheduling, interrupt or polling latency, driver and service processing, and host power-management policy when performance is declared at a host API or software measurement point.
- Virtualization, I/O translation, or resource sharing when those host mechanisms are present in the declared deployment profile.

If performance is declared only for a subset of the environmental range specified in Clause 9, that subset **shall** be explicitly stated.

When time-transfer performance is declared at a host API or software measurement point, the supplier **shall** characterize the applicable software-path latency, asymmetry, dispersion, sample age, and host-time correlation window. The declaration **shall** identify the host clock, operating-system and driver revisions, power state, load conditions, and measurement method.

6.11 Performance Documentation Checklist

The performance section of a datasheet or conformance statement **shall** include the following items when applicable:

Category	Required reporting content
Time accuracy	Reference source, measurement point, bounded error, statistic, temperature range, lock time
Time stability	MTIE and applicable TDEV intervals, measurement point, state, environmental profile
Frequency stability	Frequency accuracy, temperature stability, applicable ADEV intervals, measurement point
Phase noise	Carrier frequency, offset-frequency range, measurement bandwidth, phase-noise curve or limits
Pulse timing	Edge definition, pulse width, rise/fall time, time jitter definition, alignment to unified timescale
Holdover	Entry condition, prior lock duration, MTIE/error versus elapsed time, temperature profile
Transitions	Lock acquisition, failover, holdover exit, phase steps, reporting mechanism
Environment	Temperature, voltage, vibration, airflow, and host-integration assumptions

7. Timing Interfaces (Normative)

This clause specifies requirements for interfaces that receive or provide time, phase, frequency, and event-timestamp information. Requirements apply only to interfaces implemented or claimed by the TimeCard, except that every conforming implementation provides at least one providing interface as specified in Clause 4.

7.1 Interface classes

Timing interfaces are classified as follows:

- **Receive interfaces** acquire a reference used to initialize, discipline, validate, or monitor the unified timescale.
- **Providing interfaces** distribute a representation derived from the unified timescale.
- **Bidirectional timing interfaces** perform both functions, such as a packet interface that receives synchronization messages and provides timestamped messages.

Control operations and status associated with timing interfaces are specified in 7.4 and Clause 8.

7.2 Receive interfaces

7.2.1 Common declaration requirements

For each implemented receive interface, the supplier **shall** document:

- Interface type and direction.
- Referenced signal, protocol, and profile, including the specific edition or revision.
- Connector, pin assignment, or logical endpoint.
- Electrical or optical levels, impedance, termination, polarity, and edge definition, as applicable.
- Message format, timescale, epoch, and on-time marker, as applicable.
- Valid input range and tolerance, including amplitude, pulse width, rise and fall time, frequency offset, noise, or packet conditions relevant to acceptance.
- Measurement point and any fixed delay, asymmetry correction, or latency correction applied by the TimeCard.
- Signal-presence, validity, and loss-of-signal criteria, including hysteresis or debounce behavior.
- Acquisition, rejection, and reacquisition behavior.
- Relationship between the received reference and UTC, TAI, or another declared timescale.

7.2.2 Recognized receive-interface types

The following interface types may be implemented:

- **PTP:** If PTP is implemented, protocol behavior **shall** conform to IEEE Std 1588-2019 [4] or IEEE Std 802.1AS-2025 [5] for the profile identified in the conformance statement.
- **NTP:** If NTP is implemented as a synchronization input, protocol behavior **shall** conform to IETF RFC 5905 [7] for the mode and options identified in the conformance statement.
- **IRIG:** If an IRIG AM or DCLS input is implemented, its format **shall** conform to IRIG Standard 200-16 [8] for the code and options identified in the conformance statement.
- **GNSS:** A GNSS input may include antenna RF, receiver data, 1PPS, frequency, or a combination thereof. The implementation **shall** identify the receiver protocol, supported constellations, antenna requirements, and relationship of the receiver timescale to the unified timescale.
- **Discrete pulse or frequency:** A receive interface may accept 1PPS, another pulse rate, a sine-wave frequency reference, or a logic-level clock. Its accepted waveform and on-time or phase marker **shall** be declared as required by 7.2.1.
- **Other time-transfer methods:** A TimeCard may implement White Rabbit, regional radio time, wireless time transfer, or another method. The supplier **shall** identify the governing specification and provide the declarations required by 7.2.1.

7.2.3 Reference selection and transition behavior

If more than one receive reference is implemented, reference selection **shall** conform to 5.5. The active source, source availability, source health, and transition state **shall** be observable through at least one control interface.

If operator selection or priority configuration is implemented, the supported operations and their effect on the active source **shall** be documented. Automatic selection, manual selection, and ensemble operation **shall** be distinguishable in reported state.

For each reference-loss, failover, and reacquisition transition, the supplier **shall** declare whether the unified timescale remains continuous, the maximum phase or frequency transient, and the control indication generated by the transition.

7.2.4 1PPS receive-interface declarations

If a 1PPS receive interface is implemented, the supplier **shall** document the input impedance, valid low and high voltage ranges, pulse-width range, on-time edge, rise- and fall-time tolerance, input protection range, and loss-of-signal thresholds. The declared values apply at the input measurement point with the declared termination.

7.3 Providing interfaces

7.3.1 Common declaration requirements

For each implemented providing interface, the supplier **shall** document:

- Interface type and referenced signal, protocol, or profile.
- Connector, pin assignment, or logical endpoint.
- Timescale, epoch, encoding, and on-time or phase marker.
- Electrical or optical levels, source impedance, termination, polarity, and drive capability, as applicable.
- Measurement point, fixed delay, applied correction, and known asymmetry.
- Granularity, resolution, update rate, and rollover behavior for digital representations.
- Alignment to the unified timescale and to other providing interfaces when alignment is claimed.
- Behavior and validity indication during startup, unlocked operation, holdover, failover, time adjustment, reset, and fault.

Typical providing interfaces include pulse outputs, frequency outputs, Time of Day messages, PTP, IRIG, PCIe PTM, host-bus timestamps, and synchronous logic clocks.

7.3.2 One pulse per second providing profile

An implementation claiming the Physical Timing Output conformance profile **shall** provide at least one 1PPS output.

The 1PPS output **shall** meet the following requirements at the declared connector with a 50 ohm load:

- The connector **shall** be an SMA, TNC, or Type-N female connector, or another mechanically retained 50 ohm coaxial female connector. For conformance testing, an implementation without an SMA female connector **shall** be supplied with or specify a mechanically retained adapter to SMA female.
- The low level **shall** be between -0.3 V and 0.3 V.
- The high level **shall** be between 1.2 V and 5.5 V.
- The rising edge **shall** be the on-time mark.
- The pulse width **shall** be at least 1 microsecond and not greater than 500 milliseconds.
- The 10% to 90% rise time of the on-time edge **shall** be less than 10 nanoseconds.

The supplier **shall** document the fall time, output protection behavior, fixed delay from the unified-timescale marker, RMS time jitter or other declared short-term variation metric, bounded time variation, measurement bandwidth, and measurement uncertainty.

The supplier **shall** state the maximum alignment error between the 1PPS output and each other providing interface that claims alignment to it. For the Physical Timing Output profile, that maximum alignment error **shall** be less than 100 nanoseconds unless a different application profile defining another limit is identified in the conformance statement.

7.3.3 Protocol and signal conformance

- If an IRIG AM or DCLS output is implemented, its format **shall** conform to IRIG Standard 200-16 [8] for the code and options identified in the conformance statement.
- If PTP is implemented as a providing interface, protocol behavior **shall** conform to IEEE Std 1588-2019 [4] or IEEE Std 802.1AS-2025 [5] for the profile identified in the conformance statement.
- If PCIe PTM is implemented, PTM behavior **shall** conform to the PCI Express Base Specification, Revision 5.0, Version 1.0 [13]. The conformance statement **shall** identify the PCIe revision implemented and the evidence used to verify the Revision 5.0 PTM requirements incorporated by this standard.
- If a pulse or frequency output claims ITU-T G.703 conformance, the output **shall** conform to the applicable requirements of ITU-T Recommendation G.703 (04/2016), including Amendment 1 (05/2021) [9], at the declared measurement point.

For a providing interface that does not claim one of these referenced formats, the supplier **shall** provide a complete signal or protocol description sufficient to interoperate with the interface.

7.3.4 Alignment and continuity

Every providing interface **shall** derive its time, phase, frequency, or timestamp information from the unified timescale specified in 5.4.

The supplier **shall** declare the maximum alignment error between each time- or phase-bearing providing interface and the unified timescale. The declaration **shall** identify the measurement points, statistic or bound, operating state, environmental conditions, and any applied correction.

If an interface can produce a phase step, repeated timestamp, skipped timestamp, non-monotonic value, or invalid output during a state transition, that behavior **shall** be documented and **shall** be indicated through the interface or a control object.

7.3.5 Event timestamping

If a TimeCard provides event or packet timestamps, each timestamp **shall** derive from the unified timescale. The supplier **shall** document the capture point, timestamp format, resolution, granularity, latency, fixed correction, uncertainty, and behavior when the unified timescale is invalid or discontinuous.

If timestamps are transferred across a host bus or software boundary, the mapping **shall** identify which delays are included in the timestamp and which delays require correction by the receiving system.

7.4 Timing-interface control and status

For each implemented timing interface, at least one control interface **shall** expose:

- Presence and implemented-capability status.
- Administrative enable state, if the interface can be enabled or disabled.
- Signal or link availability.
- Validity and active-source state, when applicable.
- Loss-of-signal, format, range, or protocol alarms relevant to the interface.
- Configuration values that affect the declared timing behavior.

The semantics of these values **shall** conform to the baseline information model in 8.10 or to a documented extension as specified in 8.8. Authentication, authorization, update, and sanitization requirements are specified in 8.6 and 8.11.

7.5 Physical and logical integration

Externally accessible timing connectors **shall** be labeled or keyed so that their function and direction can be determined during installation. The supplier **shall** document connector mating requirements, termination, maximum cable assumptions used in performance declarations, and any adapter needed to reach the standardized profile connector.

For every standardized interface or profile claimed by the implementation, an extension **shall** preserve the standardized syntax, semantics, state transitions, and error behavior. An extension **shall** use a documented discovery and namespace mechanism, and a receiving implementation that does not advertise support for the extension **shall** be able to process the standardized portion without interpreting the extension.

7.6 Summary

This clause defines the declaration, signal, protocol, alignment, and state-reporting requirements needed to evaluate and integrate implemented timing interfaces. Clause 8 defines the associated control semantics, and Clause 6 defines the performance declarations for their measurement points.

8. Control Interfaces (Normative)

This clause specifies the control-plane requirements used to discover, configure, monitor, update, and manage a TimeCard. Every conforming implementation provides at least one control interface. A control interface may be local, host-integrated, out-of-band, remotely networked, or a combination of these forms.

8.1 Control-interface classes

Class	Examples	Principal use
Local hardware control	SMBus, I2C, I3C, GPIO	Board-level configuration and telemetry.
Host-integrated control	PCIe MMIO, command queues, driver APIs, USB	Host discovery, configuration, time access, and status.
Out-of-band control	IPMI, NC-SI, serial	Management independent of the host operating system.
Remote network control	REST, gRPC, Redfish, SNMP	Fleet management and remote telemetry.

An implementation may expose the same baseline information through more than one class. The semantic value of an object **shall** not change solely because a different control interface is used.

8.2 Baseline control capabilities

At least one implemented control interface **shall** provide access to the required baseline objects in 8.10.2. Collectively, the implemented control interfaces **shall** provide the following capabilities:

- Read implementation identity, revision, profile, and capability information.
- Read the unified-timescale state and an atomic time value.
- Read active and available synchronization-source information.
- Read active alarm and security-state information.
- Discover whether conditional configuration, telemetry, event, update, and sanitization operations are supported.

A writable operation is required only when the corresponding configurable feature is implemented or a conformance profile explicitly requires it.

8.3 Protocol mappings

8.3.1 SMBus, I2C, and I3C

An SMBus implementation **shall** conform to the System Management Bus Specification, Version 3.2 [14]. An I3C implementation **shall** conform to the MIPI I3C Basic specification, Version 1.2 [12].

An I2C-only implementation **shall** identify the I2C specification and revision used. For each local hardware control mapping, the supplier **shall** document addressing, bus speed, transfer size, byte order, clock-stretching behavior, timeout behavior, reset behavior, and the mapping of baseline objects.

8.3.2 PCIe configuration and MMIO

Each implemented PCIe control mapping **shall** satisfy 8.9. If PCIe PTM is implemented, the PCIe function **shall** expose the PTM capability structures required by the PCI Express Base Specification, Revision 5.0, Version 1.0 [13].

8.3.3 IPMI and NC-SI

An implementation claiming IPMI control **shall** conform to DMTF DSP0236 [1] for the IPMI version and command scope identified in the conformance statement. Vendor or P3335-specific commands **shall** use an assigned or documented extension mechanism, **shall** have identifiers unique within the applicable command namespace, and **shall** not redefine, alias, or overlap standard command identifiers.

An NC-SI mapping **shall** identify the NC-SI specification revision, package and channel discovery behavior, command set, and mapping of baseline objects.

8.3.4 REST, gRPC, SNMP, and Redfish

A network API **shall** publish a machine-readable schema or object description that identifies its version and maps operations to the baseline objects in 8.10.

An implementation claiming SNMP control **shall** identify the SNMP version and **shall** implement the message-processing, security, access-control, and management-information elements of the IETF RFC 3411 [6] framework that apply to the claimed management scope. The supplier **shall** identify the management information base modules, object identifiers, security model, and access-control model used.

REST, gRPC, and Redfish mappings **shall** document resource or service discovery, schema versioning, error responses, authentication method, and update compatibility.

8.3.5 Serial and USB

A serial or USB mapping **shall** document physical or USB class behavior, framing, command syntax or schema, encoding, flow control, timeout behavior, and the mapping of baseline objects. A human-readable command interface may be provided in addition to a machine-readable mapping.

8.4 Common mapping requirements

Each control-interface mapping **shall** define:

- A mapping name and major and minor version.
- Discovery and enumeration behavior.
- Data encoding, byte order, alignment, transfer size, and string encoding.
- Binary structure, message, or envelope lengths and version fields, when a binary ABI is used.
- Mapping of required and conditional objects.
- Object support, validity, stale-data, unavailable, and fault reporting.
- Error responses for unsupported operations, invalid values, reserved values, malformed requests, and access denial.
- Concurrency and atomicity behavior for multi-field values and configuration changes.

- Reset, restart, and persistence behavior.
- Host and device lifecycle behavior, including low-power transitions, removal, reconnection, and disposition of outstanding operations when applicable.
- Compatibility behavior for unknown objects, fields, enumeration values, and extensions.

A change that removes an object, changes its units or meaning, narrows its valid range, or changes its access semantics **shall** increment the mapping major version. A backward-compatible addition **shall** increment the mapping minor version.

A binary mapping **shall** carry a validated total length and mapping version in each request and response structure or in a transport envelope that unambiguously supplies the same information. A receiving implementation **shall** validate the transport length, declared length, and supported version before accessing any field. A short, oversized, internally inconsistent, or unsupported structure **shall** be rejected without performing the requested operation unless the mapping explicitly defines a backward-compatible trailing-field rule.

Reserved binary fields **shall** be transmitted as zero. A receiving implementation **shall** not interpret a reserved field as a capability or command. The mapping **shall** define whether a nonzero reserved field is ignored for forward compatibility or rejected as malformed.

A receiving implementation **shall** process every recognized object and field in a message. It **shall** leave an unknown optional object or field uninterpreted. When forwarding is supported, it **shall** preserve the unknown optional value. If an unknown element prevents the requested operation, the receiving implementation **shall** report an incompatibility. A receiving implementation **shall** not write a reserved value. An implementation receiving a reserved or invalid write value **shall** reject the operation without changing the prior value.

8.5 Atomic reads and configuration changes

Values representing a single logical observation **shall** be returned atomically. In particular, a timestamp split across multiple transport words **shall** use a latch, snapshot command, sequence counter, retry rule, or equivalent mechanism that prevents fields from different seconds or update cycles from being combined.

For background on transaction atomicity, see [B22].

For a configuration operation affecting timing behavior, the interface **shall** report whether the operation was accepted, when it became effective, and whether it caused or can cause a phase step, frequency transient, loss of lock, restart, or interruption of a providing interface.

Configuration writes **shall** either complete as one documented operation or report partial completion and the resulting state. A failed write **shall** not silently leave an indeterminate state or configuration.

Operations that set time, step phase, steer frequency, select a synchronization source, alter a discipline loop, or modify persistent timing calibration **shall** be serialized against other operations that can affect the same state. The mapping **shall** define the arbitration order, conflict response, and recovery behavior.

If more than one independent client can issue such operations, the mapping **shall** provide time-control ownership through an exclusive lease, an atomic claim operation, or an equivalent fail-closed mechanism. A non-owner operation **shall** be rejected without changing timing state. Ownership **shall** terminate or be revalidated according to documented rules for explicit release, client loss, timeout, reset, low-power transition, device removal, and driver or service restart. Time-control ownership **shall** not bypass authentication or authorization requirements.

8.6 Access control and security state

Every control interface **shall** document its authentication, authorization, transport protection, update authorization, audit, key-storage, and sanitization capabilities. The `SECURITY_STATE` object **shall** report the mechanisms supported and their current verified or active state. If no security mechanism is implemented for a control interface, the documentation and `SECURITY_STATE` object **shall** state `none` and **shall** identify the physical-access or deployment assumptions on which that choice depends.

Requirements for the Managed TimeCard and Secure Infrastructure TimeCard profiles are specified in 8.11. A control interface not claiming either profile may omit authentication only when the omission is declared in the conformance statement.

Security failure **shall** not be reported as normal successful completion. Authentication failures, authorization denials, firmware verification failures, and sanitization failures **shall** be distinguishable to an authorized operator.

For an interface exposing timing-affecting write operations, the authorization declaration **shall** identify the privileges required to set time, discipline phase or frequency, select a source, change persistent calibration, and update executable content. When the authorization mechanism supports distinct privileges, read-only monitoring **shall** be separable from these timing-affecting operations.

8.7 Events and telemetry

If event reporting is implemented, each event record **shall** include a timestamp or an explicit indication that valid time was unavailable, a severity, a source, an event type, and event-specific data. The event timestamp measurement point and timescale **shall** be documented.

The following state changes **shall** generate an event when the corresponding function and event reporting are implemented:

- Reference acquired, rejected, selected, degraded, or lost.
- Lock acquired or lost.
- Holdover entered or exited.
- Time step, frequency step, or discontinuity detected.
- Environmental or power threshold crossed.
- Firmware update, boot-integrity check, authentication, authorization, or sanitization completed or failed.

If telemetry streaming is implemented, the supplier **shall** document rate limits, timestamp behavior, dropped-sample indication, ordering, backpressure, and the relationship between streamed values and on-demand reads.

8.8 Extensions

An implementation may provide vendor-specific objects, commands, registers, messages, and telemetry. Extensions **shall** use a documented namespace or identifier range and **shall** not change the meaning of baseline objects.

A receiving implementation that does not advertise support for an extension **shall** be able to process the baseline object or message without interpreting that extension. Reserved P3335 identifiers and values **shall** not be assigned to vendor-specific behavior.

8.9 PCIe host interface profile

An implementation claiming the PCIe Host Mapping conformance profile **shall** expose a discoverable PCIe function associated with the TimeCard timing and control functions and a P3335 discovery descriptor satisfying 8.9.1.

The PCIe mapping **shall** document:

- Vendor ID, device ID, class code, subsystem identifiers, and revision identifier.
- PCIe Base Specification revision and discovery behavior.
- BAR, capability, command-queue, or driver-API resources used by the mapping.
- P3335 discovery-descriptor locator and serialized encoding.
- Interrupt or polling behavior.
- Function-level, fundamental, and software-reset behavior.

- Persistence of configuration and time state across each reset type.
- Active and non-active host power-state behavior, removal and reconnection behavior, and disposition of outstanding operations.
- Error reporting for unsupported, malformed, reserved, or unauthorized accesses.
- Timestamp format, epoch, timescale, granularity, rollover, and atomic-read mechanism.
- Measurement point and correction terms relating PCIe or PTM time to the unified timescale and other providing interfaces.

8.9.1 P3335 discovery descriptor

The P3335 discovery descriptor **shall** be read-only and reachable using PCI configuration-space information and a documented baseline locator. Locating the descriptor **shall** not require probing guessed MMIO offsets or accessing an optional resource.

The descriptor **shall** contain:

- A recognizable P3335 signature, descriptor revision, total length, and header length.
- Mapping major and minor version, byte order, and entry-size information.
- Implementation or board profile and active firmware or gateware image identity, including an explicit `not-applicable` or `unavailable` indication when no such image identity exists.
- The `TC_INSTANCE_ID` value or a reference to the baseline operation that returns it, together with its declared stability scope.
- BAR index, offset, and span occupied by the descriptor.
- Capability and valid-field indications sufficient to distinguish implemented, unavailable, and unsupported resources.
- Resource-directory entry count and a bounded mechanism for traversing entries.
- A generation, snapshot, or equivalent consistency mechanism by which a receiver can detect a descriptor change during discovery.

Each resource-directory entry **shall** identify the resource type and instance, BAR index, offset, span, resource revision, access modes, and applicable interrupt or polling information. An entry **shall** include its length so that an unknown optional entry type can be skipped without interpreting its fields.

Every advertised resource range **shall** fit within the system-assigned BAR length and **shall** satisfy the alignment and overlap rules declared by the mapping. A capability **shall** not be advertised solely because a register address or core revision exists; the active implementation or image **shall** provide the semantics associated with that capability.

A driver or other receiving implementation using the P3335 discovery descriptor **shall** validate the signature, supported descriptor and mapping versions, lengths, consistency mechanism, and every resource range before accessing a resource. If the implementation identity, active image, register layout, or resource range is ambiguous or cannot be validated, the receiver **shall** limit access to resources that remain unambiguously described or **shall** report the mapping as unsupported. It **shall** not infer an optional resource solely from an enumeration index, PCI identifier, core-version register, guessed address, or successful read from an undocumented MMIO location.

8.9.2 PCIe lifecycle and recovery

The PCIe mapping **shall** define behavior for entry into and exit from each supported host power state, orderly and surprise removal, hot-plug or tunneled-PCIe disconnection, reconnection, driver unload and reload, and host reboot. The definition **shall** identify which time, configuration, calibration, event, queue, interrupt, and time-control-ownership state persists.

When the PCIe function becomes inaccessible, an outstanding operation **shall** complete with a documented cancellation, unavailable, or fault response and **shall** not wait indefinitely or return stale data as current data. After reconnection, reset, or resume, a receiver **shall** revalidate the discovery descriptor before restoring access to optional resources. A timing-affecting operation **shall** not be resumed automatically unless the documented policy and revalidated time-control ownership authorize it.

8.9.3 Baseline host control mapping

The PCIe mapping **shall** expose all required baseline objects in 8.10.2 through MMIO, a command or queue interface, a driver API, or a documented combination thereof. The conformance statement **shall** identify which mapping is the interoperable baseline for the implementation.

A binary driver API used as the baseline mapping **shall** satisfy the ABI length, version, reserved-field, and compatibility requirements in 8.4. A mapping that exposes multiple TimeCards **shall** permit selection by `TC_INSTANCE_ID` and **shall** define behavior when a selector is absent, ambiguous, stale, or no longer present.

8.9.4 PCIe PTM

If PCIe PTM is implemented, the TimeCard **shall** expose the standard PTM capability structures and **shall** document the PTM measurement point and correction model.

8.10 Baseline control information model

The baseline information model defines transport-neutral object names, data semantics, access modes, and units. A protocol mapping may use registers, messages, resources, attributes, or API calls, provided that it preserves these semantics and satisfies 8.4 and 8.5.

8.10.1 Data types and object status

Type	Description
`bool`	Boolean false or true.
`uint8`, `uint16`, `uint32`, `uint64`	Unsigned integer of the indicated width.
`int64`	Signed 64-bit integer.
`enum`	One value from a defined symbolic set.
`string`	UTF-8 string with a mapping-defined maximum length.
`set`	Unordered set of defined symbolic values.
`timestamp`	Atomic structure containing seconds, nanoseconds, timescale, validity, and optional uncertainty.
`record`	Structure whose fields are defined by the object or mapping.

Access modes are `R` for read-only, `W` for write-only, and `R/W` for read/write. Every read response **shall** distinguish `valid`, `unavailable`, `unsupported`, `unspecified`, `stale`, and `fault` when those states can occur in the mapping.

For a `timestamp`, nanoseconds **shall** be in the range 0 through 999 999 999. The seconds and nanoseconds fields **shall** represent the same measurement instant and **shall** satisfy the atomic-read requirement in 8.5.

When a `record` contains conditional fields, the response **shall** provide field-level support and validity information or use a mapping-defined encoding that unambiguously distinguishes an absent, unsupported, unavailable, stale, and valid field. A zero numeric value **shall** not by itself indicate that a conditional field is absent or invalid.

Each mapping **shall** define the response when a value would exceed the representable range of its declared type. For every accumulating counter, the mapping **shall** define whether the value saturates or rolls over, its maximum or modulus, its reset conditions, and the indication used to detect rollover or lost counts. An accumulating counter **shall** not wrap without an observable indication.

8.10.2 Required baseline objects

The following objects **shall** be exposed through at least one control interface:

Object	Access	Type	Units	Semantics
`TC_MODEL`	R	string	none	Product or implementation model identifier.
`TC_INSTANCE_ID`	R	string	none	Mapping-scoped selector for one TimeCard instance. The mapping declares its derivation and stability across restart, removal, reinsertion, and host reboot.
`TC_HW_REV`	R	string	none	Hardware or implementation revision; `not-applicable` for an implementation without hardware.
`TC_FW_REV`	R	string	none	Active firmware, gateware, or software revision; `not-applicable` when none exists.
`TC_INFO_MODEL_REV`	R	string	none	P3335 information-model revision implemented.
`TC_PROFILE`	R	set	none	Conformance profiles claimed by the active configuration.
`TC_CAPS`	R	set	none	Implemented interface and feature capabilities.
`TC_STATE`	R	enum	none	Overall unified-timescale operating state.
`TC_TIMESCALE`	R	enum	none	Timescale represented by `TC_TIME` and timing telemetry.
`TC_TIME`	R	timestamp	s, ns	Atomic unified-timescale value at the declared read measurement point.
`SYNC_SRC_ACTIVE`	R	string	none	Identifier of the source currently used to discipline or validate the unified timescale; `none` when no source is active.
`SYNC_SRC_AVAIL`	R	set	none	Identifiers of synchronization sources currently available for selection or monitoring.
`ALARM_ACTIVE`	R	set	none	Active alarm identifiers; an empty set indicates no active alarm.
`SECURITY_STATE`	R	record	none	Supported and active security mechanisms and the result of the most recent integrity verification.

A required baseline object **shall** not report `unsupported` or `unspecified` as its object-level status. An optional field within that object may report `unspecified` when the mapping permits it. A required baseline object may report `unavailable`, `stale`, or `fault` only when that state is meaningful and distinguishable from a valid value.

8.10.3 Conditional baseline objects

The following objects are required when the corresponding capability is advertised in `TC_CAPS`:

Object	Access	Type	Units	Applicability and semantics
`SYNC_SRC_SELECT`	R/W	string	none	Implementations supporting operator source selection; selected source or selection policy.
`SYNC_SRC_HEALTH`	R	record	none	Implementations with receive interfaces; health and rejection reason for each source.
`PHASE_ERR`	R	int64	ps	Implementations reporting phase error; value relative to the identified source and measurement point.
`FREQ_OFFSET`	R	int64	1e-15	Implementations reporting fractional frequency offset; value relative to the identified source.
`MTIE_EST`	R	uint64	ps	Implementations reporting estimated or measured MTIE; paired with `MTIE_INTERVAL`.
`MTIE_INTERVAL`	R	uint64	s	Observation interval associated with `MTIE_EST`.
`HOLDOVER_ELAPSED`	R	uint64	s	Implementations supporting holdover; elapsed time since holdover entry.
`TEMP_LOCAL`	R	int64	mdeg C	Implementations reporting a local timing-function temperature in millidegrees Celsius; sensor location is declared.
`EVENT_COUNT`	R	uint32	count	Implementations with an event log; number of readable retained events. Behavior at the `uint32` limit and indication of event loss or overwrite follow the mapping declaration required by 8.10.1.
`EVENT_READ`	R	record	none	Implementations with an event log; next or selected event record.
`REFERENCE_EVIDENCE`	R	record	none	Implementations reporting reference-chain evidence; identifiers and availability of source, calibration, measurement-point, and uncertainty information without asserting end-to-end traceability.
`TC_SERIAL`	R	string	none	Implementations with a supplier-assigned or otherwise persistent device serial identifier; value and identity namespace are declared.
`HOST_TIME_CORRELATION`	R	record	none	Host mappings supporting bounded correlation of a TimeCard timestamp with a host clock; record satisfies 8.10.4.
`TIME_CONTROL_STATUS`	R	record	none	Host mappings supporting explicit time-control ownership; arbitration and ownership status satisfies 8.10.5.
`UPDATE_CMD`	W	record	none	Implementations supporting controlled firmware, gateware, or software update.
`SANITIZE_CMD`	W	record	none	Implementations claiming sanitization; scope and authorization parameters.

01 A conditional object advertised in `TC_CAPS` **shall** not report `unsupported`. An unadvertised conditional

object may report `unsupported` when addressed through a mapping that permits object probing.

8.10.4 Host-time correlation record

A `HOST_TIME_CORRELATION` record **shall** contain the following fields:

Field	Required	Type	Units	Semantics
`card_time`	Yes	timestamp	s, ns	Atomic unified-timescale value at the declared card measurement point.
`host_time_before`	Yes	timestamp	s, ns	Value returned by a host-clock read that completes before the card-time measurement operation begins.
`host_time_after`	Yes	timestamp	s, ns	Value returned by a host-clock read that begins after the card-time measurement operation completes.
`host_clock_id`	Yes	string	none	Stable identifier for the host clock used by both host timestamps.
`capture_sequence`	Yes	uint64	count	Monotonically advancing correlation-capture sequence within its declared reset scope.
`discontinuity_sequence`	Yes	uint64	count	Generation identifying host-clock discontinuities that invalidate prior correlations.
`correlation_window`	Yes	uint64	ns	Non-negative difference between `host_time_after` and `host_time_before`.
`sample_age`	Yes	uint64	ns	Age of the captured sample at the response or publication measurement point.
`card_state`	Yes	enum	none	`TC_STATE` applicable to the captured card timestamp.
`source_valid`	Yes	bool	none	Whether the synchronization-source evidence required by the declared host policy was valid at capture.
`discipline_eligible`	Yes	bool	none	Whether the complete record meets the declared policy for submission to a host-clock discipline function.

The two host timestamps **shall** use the same host clock, epoch, timescale, and units. The card-time measurement operation **shall** begin after the `host_time_before` read completes and **shall** complete before the `host_time_after` read begins. If that ordering cannot be established, or if the host clock is discontinuous within the interval, the record **shall** report `unavailable` or `fault` and **shall** not be marked discipline-eligible.

The mapping **shall** document the host clock, its adjustment behavior, the sign convention for any timescale correction, the maximum accepted correlation window, the maximum sample age, and the uncertainty or dispersion method used by the host integration. If the relation between the card and host timescales is not known unambiguously, `card_time` **shall** identify its timescale as `unknown` or the applicable declared value, no cross-timescale offset **shall** be asserted, and `discipline_eligible` **shall** be false.

A correlation **shall** be marked discipline-eligible only when the declared card state, source validity, timescale relation, sample age, correlation window, and uncertainty or dispersion limits are satisfied. A host-clock discontinuity **shall** advance `discontinuity_sequence` and **shall** make correlations from an earlier sequence stale or unavailable.

8.10.5 Time-control status record

A TIME_CONTROL_STATUS record **shall** identify the arbitration mode, whether ownership is active, whether the requester is the current owner, an ownership-generation value, the applicable timeout or expiry condition, and the most recent ownership-termination reason. The mapping **shall** define claim, renewal, and release operations and **shall** protect owner identity from unauthorized disclosure.

8.10.6 Common symbolic values

TC_PROFILE **shall** use the profile names base, physical-timing-output, pcie-host-mapping, managed-timecard, and secure-infrastructure-timecard for the corresponding profiles in 4.4.

TC_STATE **shall** distinguish at least the following states: unknown, initializing, warming-up, free-running, acquiring, locked, holdover, degraded, and fault.

TC_TIMESCALE **shall** distinguish at least unknown, TAI, UTC, PTP, and other-declared. If other-declared is used, the mapping to a declared reference timescale **shall** be documented.

Unknown symbolic values received by a receiving implementation **shall** be preserved when forwarding and otherwise treated as unknown. A mapping that encodes symbolic values numerically **shall** document the numeric assignments and reserve an extension range.

8.10.7 GNSS conditional objects

If a GNSS receive capability is advertised in TC_CAPS, the following objects **shall** be exposed:

Object	Access	Type	Units	Semantics
GNSS_INSTANCES	R	uint8	count	Number of GNSS receiver instances.
GNSS_ENABLE	R/W	set	none	Enabled receiver-instance identifiers, when receiver enable control is supported.
GNSS_CONST_AVAIL	R	record	none	Available constellations for each receiver instance.
GNSS_CONST_ENABLE	R/W	record	none	Enabled constellations for each receiver instance, when configuration is supported.
GNSS_LOCK_STATE	R	record	none	Acquisition or lock state for each receiver instance.
GNSS_ANTENNA_STATE	R	record	none	Antenna state for each receiver instance, including unknown, valid, open, short, or degraded when detectable.

The common constellation identifiers are GPS, Galileo, GLONASS, BeiDou, QZSS, and NavIC. Additional constellation identifiers **shall** use the extension mechanism defined by the control mapping.

8.10.8 Other satellite timing sources

Non-GNSS satellite services, including large low-Earth-orbit constellations such as Starlink, are not assigned common GNSS identifiers by P3335. If an implementation uses such a service as a timing source, the supplier **shall** identify the service, document vehicle selection and handover behavior, and expose source identity and health through the extension mechanism defined by the control mapping.

8.11 Security profiles

8.11.1 Baseline security declaration

All conforming implementations **shall** document the security properties of each control interface, including physical-access assumptions. The documentation **shall** state whether authentication, authorization, encrypted transport, signed update, boot integrity verification, protected key storage, anti-rollback protection, event auditing, tamper response, and sanitization are supported. This declaration requirement does not require a security mechanism unless the implementation claims a profile or feature whose applicable requirements specify that mechanism. When no security mechanism is implemented, the documentation **shall** state `none` and identify the deployment assumptions.

8.11.2 Managed TimeCard profile

An implementation claiming the Managed TimeCard profile **shall** meet the following requirements:

- A networked or remotely reachable control interface **shall** authenticate the user or machine before permitting configuration or update operations.
- Authorization **shall** distinguish monitoring, time setting and discipline, configuration, update, security-administration, and sanitization privileges when those operations are implemented.
- Default shared credentials **shall** be replaced or changed before remote configuration is enabled.
- An update payload **shall** be integrity-checked before installation.
- Security-relevant successes and failures **shall** be recorded when event logging is implemented.
- The security protocol, version, credential lifecycle, and recovery procedure **shall** be documented.

8.11.3 Secure Infrastructure TimeCard profile

An implementation claiming the Secure Infrastructure TimeCard profile **shall** meet the Managed TimeCard profile and the following additional requirements:

- Firmware, gateware, and other executable update payloads **shall** be digitally signed and verified against an authorized trust anchor before activation.
- The update mechanism **shall** prevent installation of an unauthorized rollback version unless an explicitly authorized recovery procedure is used.
- Secure boot or measured boot **shall** verify executable integrity before the timing service enters the locked state.
- Private keys and credentials **shall** be stored in hardware-backed or equivalently protected storage whose protection mechanism is documented.
- Remote IP-based management **shall** use an encrypted, peer-authenticated transport.
- Privileged operations **shall** provide replay and message-injection protection.
- Sanitization **shall** erase the sensitive keys, credentials, and security-relevant configuration identified in the sanitization scope.
- Sanitization completion or failure **shall** be reported, and a failed sanitization **shall** not be reported as successful.

8.12 Summary

This clause defines a transport-neutral baseline control model and the requirements that each protocol binding preserves. The model supports common integration while allowing transport-specific and vendor-specific extensions to evolve through explicit versioning and namespaces.

9. Environmental Specifications (Normative)

This clause specifies environmental, mechanical, electrical, qualification, and lifecycle information needed to evaluate a TimeCard and its declared timing performance. It uses a declaration model because the applicable limits depend on implementation form and deployment environment.

9.1 Applicability

A conforming implementation **shall** identify its intended deployment environment and implementation form. Physical TimeCards **shall** provide the declarations applicable to their enclosure, board, connectors, power, cooling, and service conditions. Embedded or virtualized implementations **shall** identify the host-provided environmental and resource conditions on which their declared performance depends.

An environmental declaration **shall** distinguish:

- **Operating limits**, within which the implementation is functional.
- **Full-performance limits**, within which the Clause 6 performance declarations apply.
- **Storage and transport limits**, within which the unpowered implementation is not permanently damaged or degraded.
- **Survival limits**, if claimed, after which operation can require inspection, recalibration, or repair.

9.2 Operating and storage conditions

The supplier **shall** declare the conditions and elapsed interval after startup at which normal operation begins, even when full-performance limits do not yet apply. The declaration **shall** distinguish first usable output, normal operation, locked operation, and full-performance operation when those milestones differ.

9.2.1 Temperature

The supplier **shall** declare the operating, full-performance, and storage temperature ranges that apply at identified measurement locations. The declaration **shall** state whether the temperature is ambient, inlet-air, component, case, junction, or another defined quantity.

The supplier **shall** declare warm-up conditions and any thermal stabilization interval needed before full-performance limits apply. Temperature coefficients or bounded timing deviations **shall** be reported as required by 6.10 and 9.5.

9.2.2 Humidity and condensation

For a physical implementation, the supplier **shall** declare the operating and storage relative-humidity ranges and whether the limits are non-condensing. If condensation, conformal coating, sealing, or corrosion protection affects use or recovery, the relevant condition and procedure **shall** be documented.

9.2.3 Altitude, pressure, and cooling

If altitude or ambient pressure affects cooling, electrical spacing, oscillator behavior, or declared performance, the supplier **shall** declare the applicable altitude or pressure range and any derating.

The supplier **shall** document cooling assumptions needed for safe operation and full performance, including airflow direction and minimum airflow, heat-sink or cold-plate requirements, allowable inlet temperature, or host thermal design requirements, as applicable.

Where forced air, liquid, or another coolant is required, the supplier **shall** document the coolant type, cleanliness or contamination limits, filtration requirements, inspection or maintenance interval, and any derating or protective response required when a filter, heat sink, cold plate, or other heat-transfer surface becomes obstructed or fouled.

9.3 Mechanical conditions

9.3.1 Form factor and installation

A physical implementation **shall** document dimensions, mass, mounting points, connector locations, installation orientation, keep-out areas, insertion or extraction constraints, and retention requirements. If a standardized form factor is claimed, the conformance statement **shall** identify the governing specification and revision.

9.3.2 Shock and vibration

If operation, survival, or performance under shock or vibration is claimed, the supplier **shall** declare the test profile, axes, mounting fixture, operating state, sample quantity, acceptance criteria, test method, and governing standard and edition, if any. Frequency sensitivity to vibration **shall** be characterized using IEEE Std 1193-2022 [3] or another method identified in the conformance statement.

9.3.3 Connectors and cabling

The supplier **shall** document mating connectors, retention or torque requirements, the expected number of mating cycles where relevant, cable strain-relief assumptions, and environmental limitations for externally accessible timing and RF connectors.

9.4 Electrical and electromagnetic conditions

9.4.1 Power

For each power source, the supplier **shall** document nominal voltage, operating tolerance, maximum steady-state power, startup or inrush behavior, sequencing, auxiliary or standby power, and behavior outside the operating range.

An implementation powered through a standardized host interface **shall** conform to the power, sequencing, reset, and current limits of the identified host-interface specification. The supplier **shall** document which state, configuration, time, and calibration information persists through each power interruption or reset condition.

For a host-connected implementation, the supplier **shall** document whether the TimeCard remains powered, maintains the unified timescale, or retains time-control ownership while the host enters a low-power, sleep, hibernation, shutdown, or disconnected state. The documentation **shall** identify the timing-service and control-interface behavior during transition and recovery.

9.4.2 Supply sensitivity and interruption

The effect of input-voltage variation and supply noise on declared timing performance **shall** be characterized when applicable. If ride-through, backup energy, or uninterrupted holdover is claimed, the supplier **shall** declare the interruption profile, supported duration, load condition, output behavior, and recovery behavior.

9.4.3 EMC and ESD

For a physical implementation, the supplier **shall** identify each EMC emission, immunity, and ESD standard for which conformity is claimed, including edition, test level, port classification, configuration, and result. P3335 conformance does not by itself constitute regulatory approval or EMC/ESD certification.

The supplier **shall** document any operating restriction, cable, enclosure, grounding, or shielding condition required for a claimed EMC, ESD, or timing-performance result.

9.5 Environmental performance characterization

Environmental measurements supporting a conformance claim **shall** satisfy the measurement requirements in 6.3. For each environmental factor evaluated, the report **shall** identify:

- Applied stimulus, range, rate of change, dwell, and sequence.
- TimeCard operating state, active reference, and configuration.
- Measurement point and monitored timing metric.
- Test equipment, calibration status, and measurement uncertainty.
- Number of specimens and preconditioning.
- Performance limit and pass/fail rule.
- Pre-test, during-test, and post-test results.

- Recovery or recalibration performed after the test.

Within the declared full-performance range, measured performance **shall** remain within the applicable bounded declarations from Clause 6. Within the wider operating range, the implementation **shall** remain functional. When the implementation can detect that the applicable full-performance bounds are not assured, it **shall** expose a dynamic indication through a control object or alarm, and the supplier **shall** document the indication, the affected bounds, and the conditions for setting and clearing it. When the implementation cannot detect that condition, the supplier **shall** document the external monitoring needed to determine whether the bounds apply.

9.6 Reliability, service life, and calibration

9.6.1 Reliability declarations

If MTBF, failure rate, availability, or another reliability metric is claimed, the supplier **shall** document the metric definition, prediction or field-data method, environment, duty cycle, confidence basis, exclusions, and source data revision. A predicted reliability value **shall** be distinguishable from a field-observed value.

9.6.2 Service-life items

The supplier **shall** identify components or stored data with a service-life limitation that can affect timing service, including batteries, fans, nonvolatile-memory endurance, oscillators, and calibration data, as applicable. The documentation **shall** state the rated interval or endurance, replacement or maintenance action, and effect of expiration or failure.

9.6.3 Calibration and traceability

If calibration is required to maintain a declared performance bound, the supplier **shall** document the calibration interval or triggering condition, measurement point, reference, procedure, adjustable parameters, and post-calibration verification. Calibration records **shall** identify the implementation revision, date, result, uncertainty, and traceability chain.

9.7 Environmental documentation checklist

The conformance statement or referenced product documentation **shall** include, as applicable:

Category	Required content
Deployment	Intended environment and implementation form.
Temperature	Operating, full-performance, storage, and survival ranges; measurement location; warm-up.
Humidity	Operating and storage ranges; condensation condition.
Altitude and cooling	Range, derating, airflow direction and rate, coolant cleanliness, filtration and maintenance assumptions, or other cooling dependency.
Mechanical	Form factor, dimensions, mass, mounting, retention, shock, and vibration claims.
Power	Rails, tolerances, consumption, inrush, sequencing, reset, persistence, and interruption behavior.
EMC and ESD	Claimed standards, editions, levels, configurations, restrictions, and results.
Performance	Environmental characterization method, uncertainty, limits, and qualification evidence.
Lifecycle	Reliability basis, service-life items, maintenance, and calibration.

9.8 Regulatory note (Informative)

Manufacturers and integrators remain responsible for the safety, electromagnetic, environmental, spectrum, and other legal requirements of the jurisdictions in which a product is produced, installed, or operated. P3335 only defines TimeCard conformance and does not and cannot grant regulatory approval.

10. Applications and Best Practices (Informative)

This clause provides deployment considerations for TimeCard systems. It does not add or alter conformance requirements. Application-specific accuracy, availability, security, safety, and regulatory limits remain the responsibility of the system designer and operator.

10.1 Selecting a TimeCard

A useful selection process begins with the system timing requirement and works back to the TimeCard measurement point. The evaluation should consider:

- Required timescale and traceability path.
- Maximum permitted time error, frequency offset, or phase error at the consuming application.
- Network, host-bus, software, cable, and interface error between the TimeCard measurement point and that application.
- Required availability and maximum duration of reference loss.
- Environmental and host-integration conditions.
- Monitoring, update, access-control, and recovery needs.

Timestamp granularity or oscillator type alone does not establish end-to-end accuracy. The complete error and uncertainty path and budget should be evaluated.

10.2 Application considerations

10.2.1 Data centers and distributed computing

TimeCards can serve as PTP grandmasters, host timing sources, or monitored references. Deployments commonly consider redundant references, independent failure domains, hardware timestamping, host PHC integration, monitoring at scale, and the effect of network asymmetry. The relevant accuracy target should be set by the application rather than inferred from the interface technology.

10.2.2 Telecommunications

Telecommunications deployments can require specific PTP or frequency-synchronization profiles and bounded holdover during reference failure. Source traceability, packet-network support, asymmetry, environmental qualification, and profile-specific limits should be evaluated together.

10.2.3 Regulated event timestamping

Financial, industrial, and other regulated systems can require demonstrable traceability, bounded divergence from UTC, retained evidence, and controlled configuration. Operators should map the applicable rule to the actual event-timestamp measurement point and include all distribution and capture delays in the uncertainty analysis.

10.2.4 Power and industrial systems

Industrial and power-system deployments can combine precise timing requirements with extended environmental, EMC, safety, and availability constraints. Applicable industry profiles, qualification standards, isolation, grounding, and fail-safe behavior should be identified by the system specification.

10.2.5 Scientific and metrology systems

Scientific applications can prioritize phase coherence, low phase noise, long observation intervals, or calibration access. The selected metrics and test setup should match the experiment's sensitivity and should preserve raw observations needed for later analysis.

10.3 Installation

10.3.1 Thermal and mechanical integration

The host should provide the airflow, inlet temperature, orientation, and mounting conditions used by the supplier's full-performance declaration. Large local heat sources and changing fan policies can introduce thermal gradients and timing effects that vary independently of ambient temperature.

10.3.2 Cabling and delay

Cable type, length, temperature coefficient, connector condition, termination, and adapter delay can materially affect a physical timing measurement point. Fixed corrections should use a consistent sign convention and should be recorded with uncertainty. Installation changes, including connector replacement, cable rerouting, adapter changes, or altered strain relief, should trigger review or recalibration.

10.3.3 Antenna and RF paths

GNSS antenna placement should provide the required sky view and should account for cable loss, antenna power, lightning protection, RF interference, jamming, spoofing, and correlated-failure risks. Antenna delay and receiver configuration should be included in the declared reference path.

10.3.4 Host software and driver integration

Host software should begin with the declared mapping and capability information, validate every resource range, and keep optional resources inaccessible until the active implementation or image establishes their semantics. Successful reads from guessed MMIO locations, PCI identifiers, or core-version values alone are not reliable evidence that an optional feature is implemented.

Multi-card software should select a card using `TC_INSTANCE_ID` or `TC_SERIAL` when available. An enumeration index or operating-system device number can change after removal, reinsertion, firmware update, or inventory changes and should not be used as the persistent key for configuration or calibration data.

Software correlating TimeCard time with a host clock should preserve both host bounds, the correlation window, capture and discontinuity sequences, card state, source validity, sample age, and the card and host timescales. A raw hardware counter should not be labeled UTC, or used to discipline a host clock, until the timescale relationship and applicable leap information are valid.

Only one authorized component should control time setting or discipline at a time. Monitoring applications, command-line tools, background discipline services, and host time providers should share the ownership and authorization mechanism defined by the mapping rather than applying independent steering actions.

Sleep, wake, hibernation, driver restart, orderly removal, surprise removal, and tunneled-PCIe disconnection should be treated as normal lifecycle cases. Software should cancel bounded operations, stop hardware access when resources disappear, revalidate discovery after recovery, and reject samples captured before an intervening host-clock discontinuity.

Annex E gives platform-specific examples for Windows, macOS, and Linux.

10.4 Operation and maintenance

10.4.1 Reference policy

Reference priorities and qualification thresholds should reflect independence, accuracy, stability, and failure behavior. A secondary source that shares an antenna, power supply, network path, or upstream clock with the primary source might not provide meaningful redundancy.

10.4.2 Monitoring and alarms

Operators should monitor at least source availability, active source, lock state, holdover elapsed time, phase error, frequency offset, environmental conditions, alarms, and software revisions. Alert thresholds should be tied to the time remaining before an application limit might be exceeded, not only to device state names.

10.4.3 Updates and configuration

Configuration and update changes should be staged, authorized, recorded, and evaluated for their effect on time continuity. A rollback plan should account for configuration compatibility, security policy, calibration data, and the possibility that timing service is interrupted during recovery.

10.4.4 Calibration and evidence retention

Calibration and verification intervals should be based on the declared performance, observed aging, environmental exposure, and application risk. Records should preserve the device identity, configuration, reference chain, uncertainty, result, and processing method.

10.5 Common failure syndromes

Symptom	Possible cause	Investigation
Constant offset	Cable or antenna delay, wrong edge, epoch mismatch, or unapplied correction	Trace the measurement point, correction sign, and correction value through the complete timing path.
Intermittent source loss	Marginal signal level, RF interference, packet loss, threshold hysteresis, loose or poorly retained connectors, broken or intermittent cabling, or lack of needed redundancy	Inspect and mechanically verify connectors and cabling, then correlate source health, raw signal indicators, network data, and event logs.
Excessive holdover error	Initial frequency offset, temperature change, aging, insufficient preconditioning, or model error	Compare the holdover test conditions with the supplier declaration and recorded environment.
Host timestamp disagreement	Different epochs or timescales, non-atomic reads, host-bus delay, or conversion error	Validate the timestamp mapping, rollover behavior, atomicity, and correction model.
Unexpected phase step	Reference switch, manual adjustment, restart, or invalid continuity assumption	Correlate physical outputs with control state and transition events.
Management loss during host fault	Control path shares host power, software, or network dependencies	Review the intended out-of-band boundary and correlated-failure assumptions.

Annex A: Metrics (Informative)

This annex summarizes timing and frequency metrics used by the normative performance declarations in Clause 6. The controlling definitions and calculation methods are those in the normative references cited by the applicable requirement. The environmental-sensitivity discussion in A.10 condenses concepts from IEEE Std 1193-2022 [3] for convenient application to TimeCards; IEEE Std 1193-2022 remains the controlling source where a normative requirement cites it.

A.1 Metric selection

No single metric characterizes every aspect of a TimeCard. A useful declaration selects metrics according to the behavior being evaluated:

Behavior	Common metric or result
Offset from a reference timescale	Time error with a stated bound or statistic
Bounded time variation	MTIE
Stochastic time variation	TDEV
Fractional-frequency stability	ADEV or modified ADEV
Spectral purity of a periodic output	Phase noise
Short-term edge variation	A specifically defined time-jitter statistic
Reference-loss behavior	Time error or MTIE versus elapsed holdover time

The metric name alone is insufficient. The measurement point, operating state, averaging or observation interval, bandwidth, environmental conditions, and uncertainty determine what a result means and therefore should be documented with the result.

A.2 Allan deviation

Allan deviation (ADEV) characterizes fractional-frequency stability as a function of averaging time τ. For N successive fractional-frequency averages $\bar{y}_i$ of duration τ, the non-overlapping estimator is:

$$\sigma_y(\tau) = \sqrt{\frac{1}{2(N-1)} \sum_{i=1}^{N-1} \left(\bar{y}_{i+1} - \bar{y}_i\right)^2}.$$

Overlapping estimators use more of the available data and can provide better confidence, but the estimator, sampling interval, dead time, preprocessing, and confidence basis should be reported. ADEV values from different averaging intervals or estimator configurations are not directly interchangeable.

A.3 Time deviation

Time deviation (TDEV) characterizes stochastic time variation as a function of averaging time. It is related to modified Allan deviation $\operatorname{mod}\sigma_y(\tau)$ by:

$$\mathrm{TDEV}(\tau) = \frac{\tau}{\sqrt{3}} \operatorname{mod}\sigma_y(\tau).$$

TDEV is useful when assessing time-transfer noise and wander. A report should identify the underlying time-error sequence, sampling interval, averaging factors, filtering, and treatment of gaps or outliers.

A.4 Maximum time interval error

Maximum time interval error (MTIE) is the maximum peak-to-peak time-error variation in any observation window of duration τ. For a time-error function $x(t)$ over the available record, it can be expressed as:

$$\mathrm{MTIE}(\tau) = \max_{t_0} \left[\max_{t_0 \le t \le t_0+\tau} x(t) - \min_{t_0 \le t \le t_0+\tau} x(t) \right].$$

MTIE is a worst-observed-window statistic. The total record length, sampling interval, treatment of missing data, and observation intervals should accompany a reported curve or limit.

A.5 Phase noise

Single-sideband phase noise $\mathcal{L}(f)$ describes noise power in a 1 Hz bandwidth at offset frequency f from a carrier, relative to carrier power, and is normally reported in dBc/Hz. A declaration should identify carrier frequency, offset-frequency range, resolution and measurement bandwidths, cross-correlation settings if used, instrument floor, and any spurs excluded from the noise trace.

Phase noise is a frequency-domain metric. It should not be substituted for a time-domain jitter, ADEV, TDEV, or MTIE result without a stated conversion method and integration limits.

A.6 Time jitter

The word *jitter* has several domain-specific meanings. For a pulse or event stream, a TimeCard declaration should define:

- The ideal event model and any trend or deterministic component removed.
- The event or edge used as the measurement marker.
- Measurement bandwidth and observation interval.
- Sample count and handling of missing or invalid events.
- Statistic, such as RMS, standard deviation, peak-to-peak observed range, percentile, or bounded maximum.
- Whether fixed offset is included or removed.

Peak-to-peak observed range depends strongly on sample count and observation time. RMS and standard deviation are equivalent only under the stated centering convention. These details are therefore part of the metric definition.

A.7 Accuracy, precision, resolution, and granularity

Term	Use in IEEE P3335
Accuracy	Qualitative closeness to a reference value; numerical claims are expressed using a defined error and uncertainty.
Precision	Closeness of agreement among repeated indications under stated conditions.
Resolution	Smallest change in a measured quantity that produces a perceptible change in indication.
Granularity	Smallest step representable by an encoding or interface.

A fine timestamp granularity does not by itself imply equally fine resolution, precision, or accuracy.

These terms define P3335 performance-declaration semantics and are aligned with the corresponding measurement concepts used by IEEE Std 1588-2019 [4]. When an implementation claims an IEEE 1588 mapping, IEEE Std 1588-2019 governs the protocol fields and state-machine behavior, while the definitions in Clause 3 govern P3335 conformance and performance declarations. Under 4.7 and 7.3.3, an implementation-specific departure is identified in the conformance statement and is not to be represented or interpreted as IEEE 1588 conformance.

P3335 uses *accuracy* qualitatively and requires a quantitative claim to be expressed as a defined error or uncertainty bound. An IEEE 1588 `clockAccuracy` value is a protocol encoding that is mapped to the applicable measured or declared range under 6.4. P3335 uses *precision* for repeatability, *resolution* for the smallest perceptible change in indication, and *granularity* for the smallest representable encoding step. These four terms are not interchangeable, and an IEEE 1588 timestamp field width or scaling does not, by itself, establish P3335 resolution, precision, or accuracy.

A.8 Holdover time error

During holdover, time error can be represented conceptually as:

$$x_H(t) = x_0 + y_0 t + \frac{1}{2} D t^2 + \int_0^t n_y(u)\,du,$$

where x_0 is time error at holdover entry, y_0 is initial fractional-frequency offset, D is a linear fractional-frequency drift rate, and n_y represents residual frequency fluctuations. Temperature, aging, retrace, supply sensitivity, and the quality of the pre-holdover estimate can alter these terms.

A holdover declaration should therefore identify the prior lock duration and source, entry condition, elapsed holdover interval, temperature profile, initial error treatment, and behavior on reference restoration.

A.9 Ensemble and correlation considerations

For N independent sources with equal variance, ideal averaging reduces standard deviation by a factor of $\sqrt{N}$, giving an ensemble deviation proportional to $1/\sqrt{N}$. Real sources can share environmental, receiver, antenna, power, or network errors, so error correlation reduces or eliminates that benefit.

An ensemble result should identify the source population, weighting or selection method, error-correlation assumptions, rejection criteria, and behavior as sources become unavailable or degraded.

A.10 Environmental sensitivity

Common environmental sensitivity quantities include fractional-frequency change per degree of temperature, fractional-frequency change per unit acceleration, supply sensitivity, and aging rate. Reports should distinguish static sensitivity from transient response and should identify axis, stimulus frequency, rate of change, settling time, and hysteresis when relevant.

Concurrent logging of environmental quantities and timing error helps separate environmental sensitivity from reference or measurement-system noise.

Annex B: Test Procedures (Informative)

This annex provides example procedures for evaluating TimeCard functions and declared performance. It does not define additional conformance requirements. The applicable normative clause and supplier declaration provide the pass/fail limit.

B.1 Test record structure

Each test record should identify:

- Test identifier and applicable clause, profile, interface, or optional feature.
- Device under test, hardware revision, firmware or gateware revision, and configuration.
- Entrance conditions, including warm-up, lock state, source, power, and environment.

- Equipment, calibration status, traceability chain, and measurement uncertainty.
- Cabling, termination, corrections, and measurement points.
- Procedure and data-processing method.
- Declared limit, decision rule, and treatment of uncertainty.
- Result, retained evidence, and any deviation from the planned method.

A representative arrangement is shown in Figure 3.

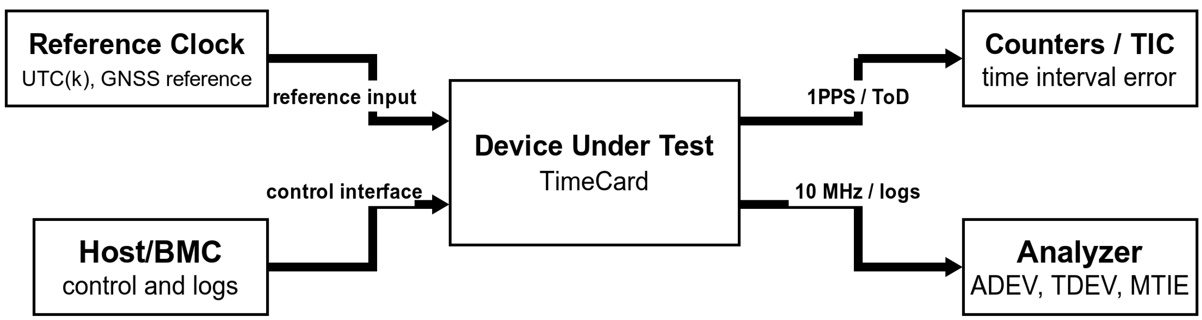

Figure 3: Representative TimeCard test fixture

B.2 General measurement conditions

The environment should be controlled and recorded at a level appropriate to the declared limit. The TimeCard should complete the supplier-declared warm-up and stabilization interval before a full-performance measurement begins.

The measurement system should have sufficient bandwidth, noise floor, resolution, stability, and uncertainty to distinguish the declared limit. A fixed instrument-to-device accuracy ratio is not assumed; the test report should justify the selected system and decision rule.

Cable delay, connector adapters, splitters, terminations, and reference-distribution paths should be characterized when their contribution is material. Corrections should be stated with sign convention and uncertainty.

B.3 Functional verification

B.3.1 Receive interfaces

Test	Applicability	Method	Example pass/fail basis
Discovery and declaration	Each receive interface	Inspect documentation and control capabilities, then compare them with the physical or logical interface.	Type, direction, protocol or signal, limits, measurement point, and source identifier are consistent.
Reference qualification and status	Each receive interface	Apply supplier-declared acceptable conditions and selected unacceptable counterexamples, then observe state.	Reported acceptance, rejection, source identity, and alarms agree with the declared qualification rules for the tested conditions.

Test	Applicability	Method	Example pass/fail basis
Boundary behavior	Physical receive interfaces	Exercise declared amplitude, frequency, pulse, or protocol acceptance boundaries without exceeding protection limits.	Accepted and rejected conditions agree with the declared thresholds and hysteresis.
Loss and reacquisition	Each receive interface used for synchronization	Remove or invalidate the active reference, then restore it.	State, alarms, transition behavior, and reacquisition time agree with the declaration.
Source selection	Multiple-reference implementations	Exercise automatic and operator-controlled selection policies.	Active source and selection mode agree with the documented policy.

P3335 does not prescribe a universal test that proves a reference valid under every possible condition. Reference qualification is evaluated using a finite, declared set of acceptable conditions and counterexamples. The test record should identify that set, the implemented qualification rules, and the limits of the resulting evidence.

B.3.2 Providing interfaces

Test	Applicability	Method	Example pass/fail basis
1PPS electrical profile	Physical Timing Output profile	Measure low level, high level, on-time edge, pulse width, and rise time into 50 ohms at the declared connector.	Every measured characteristic satisfies 7.3.2 after applying the stated decision rule.
1PPS alignment	Physical Timing Output profile	Compare the 1PPS measurement point with the unified-timescale reference and each interface claiming alignment.	Error remains within the applicable declared and profile limits.
Format and epoch	Digital time output	Decode values across startup, normal operation, and a relevant boundary such as second rollover.	Format, epoch, timescale, validity, and rollover agree with the declaration.
Discontinuity indication	Interface that can step or become invalid	Cause a declared failover, manual adjustment, reset, or fault.	Output behavior and validity indication agree with 7.3.4 and the supplier declaration.
PTM behavior	PCIe Host Mapping profile with PTM	Negotiate PTM and compare its represented measurement point with the unified timescale.	Capability discovery, timestamps, correction model, and uncertainty agree with the declaration.

B.3.3 Control interfaces

Test	Applicability	Method	Example pass/fail basis
Required objects	All implementations	Read every object in 8.10.2 through the declared baseline control mapping.	Every object is present and does not report unsupported.

Test	Applicability	Method	Example pass/fail basis
Atomic time read	All implementations	Repeatedly read `TC_TIME`, emphasizing second rollover and concurrent updates.	No result combines fields from different measurement instants; nanoseconds remain in range.
Conditional discovery	Implementations advertising conditional capabilities	Compare `TC_CAPS` with readable objects and exercised operations.	Every advertised object is mapped and unadvertised objects follow the documented unsupported behavior.
Invalid and reserved write	Writable mappings	Submit an out-of-range or reserved value in a non-destructive configuration.	The write is rejected and the prior value remains unchanged.
Reset and persistence	Reset-capable mappings	Exercise each documented reset type.	Discovery, state, persisted configuration, and time behavior agree with the mapping declaration.
Access control	Managed or Secure Infrastructure profile	Exercise monitoring, configuration, update, and security roles with authorized and unauthorized identities.	Each operation is allowed or denied according to the claimed profile and is reported unambiguously.

01 B.3.4 Host and driver mappings

Test	Applicability	Method	Example pass/fail basis
Discovery descriptor	PCIe Host Mapping profile	Locate the descriptor using only the declared baseline locator; validate signature, version, lengths, consistency generation, and every advertised range.	The descriptor is found without guessed MMIO probing, and every resource lies within an assigned BAR and matches its declared capability.
Ambiguous or unknown implementation	PCIe Host Mapping profile	Present an unsupported mapping version, inconsistent image identity, out-of-range entry, unknown optional entry, and unrecognized board profile using a fixture or controlled fault injection.	Baseline access is limited to unambiguously valid resources; optional access is rejected without an undocumented MMIO transaction.
Binary ABI framing	Binary host mappings	Exercise exact, shorter, longer, unsupported-version, inconsistent-length, and nonzero-reserved-field requests.	Accepted and rejected cases match the documented compatibility rules, and rejected requests do not change device state.
Multi-card identity	Host mappings supporting multiple TimeCards	Enumerate at least two instances, record identifiers, reorder or remove them, restart the driver, and reinsert a card.	Selection remains unambiguous; persistent state follows the declared stable identity rather than an enumeration index.

Test	Applicability	Method	Example pass/fail basis
Host-time correlation	Mappings advertising host-time correlation	Capture repeated records under normal load and at a second boundary; independently instrument the host calls where practical.	The card read is bounded by the two host readings, sequences advance as declared, the correlation window is consistent, and all timescales and measurement points are identified.
Timescale and eligibility safety	Mappings offering samples for host discipline	Invalidate source state, timescale correction, freshness, correlation-window, and uncertainty inputs one at a time. Cause a host-clock discontinuity.	Ineligible or pre-discontinuity samples are reported stale or unavailable and are not submitted as valid discipline samples.
Time-control ownership	Host mappings permitting multiple timing-control clients	Race two authorized clients for ownership; exercise non-owner writes, timeout, client termination, reset, sleep, and removal.	At most one client controls timing state; non-owner writes have no effect; release and recovery follow the declared rules.
Power and removal lifecycle	Host-connected implementations	Exercise supported sleep, hibernation, wake, orderly removal, surprise removal or tunnel disconnect, reinsertion, and driver reload while reads, events, and a bounded write are outstanding.	Operations complete or cancel within declared bounds, no stale sample is returned as current, resources are not accessed while absent, and discovery is revalidated before optional access resumes.

B.4 Performance measurements

B.4.1 Time accuracy

Compare the declared output or timestamp measurement point with the declared reference timescale. Record enough data to support the stated maximum, percentile, RMS, or confidence result. Include fixed-delay corrections, source-to-reference uncertainty, and measurement-system uncertainty.

B.4.2 ADEV and TDEV

Acquire a continuous phase or time-error record for a duration sufficient to support the declared averaging intervals. State the sample interval, estimator, overlap, detrending, gap handling, and confidence basis. Compare the resulting curve or points with the bounded declaration.

B.4.3 MTIE

Acquire the time-error sequence at the declared measurement point and compute MTIE for the stated observation intervals. Record total data length, sample interval, window implementation, missing-data treatment, and preprocessing. Compare each applicable interval with the declared bound.

B.4.4 Phase noise

Measure the declared carrier with an analyzer configuration whose residual floor is characterized. Record offset range, resolution and measurement bandwidths, cross-correlation settings, averaging, spurs, and instrument floor. Compare the trace with the declared mask or points.

B.4.5 Pulse timing variation

Capture the declared edge using the stated termination and bandwidth. Use the declared sample population and ideal-event model. Compute the exact RMS, standard-deviation, percentile, peak-to-peak, or bounded statistic stated by the supplier.

B.4.6 Holdover

Lock the TimeCard for at least the declared preconditioning interval, record initial time and frequency error, remove all applicable references, and observe the declared holdover duration. Record the temperature and power profile. Compare time error or MTIE versus elapsed holdover time with the supplier's bound, then verify reference-restoration behavior.

B.5 Environmental and transition tests

Environmental tests should follow the supplier-declared profile and the reporting fields in 9.5. Timing measurements should continue during the applied stimulus when the claim concerns operational performance, rather than relying only on a post-test functional check.

Transition testing should cover each state change identified in 6.9. The record should correlate physical outputs, digital timestamps, host-time correlations, control state, ownership, alarms, events, and host lifecycle state so that continuity and reporting behavior can be evaluated against the same timeline.

B.6 Reporting

The final report should include raw or losslessly transformed data, processing software and version, configuration files, plots with units and uncertainty, and a clear mapping from every conclusion to the applicable requirement and declared limit. When data cannot be shared, the report should identify the retained evidence and access authority.

Annex C: Bibliography (Informative)

This bibliography identifies background material useful for designing, applying, and evaluating TimeCard implementations. These documents are informative. Their implementation is not required for conformance unless a normative clause separately cites a document listed in Clause 2.

C.1 Standards and Industry Profiles

- **[B1]** CENELEC EN 55032:2015+A11:2020 and EN 55035:2017+A11:2020, *Electromagnetic compatibility of multimedia equipment–Emission Requirements* and *Immunity Requirements*.
- **[B2]** ETSI EN 300 019 series, *Environmental Engineering (EE); Environmental conditions and environmental tests for telecommunications equipment*.
- **[B3]** IEC 60068-2-6:2007, *Environmental testing–Part 2-6: Tests–Test Fc: Vibration (sinusoidal)*. [https://webstore.iec.ch/en/publication/544]
- **[B4]** IEC 60068-2-27:2008, *Environmental testing–Part 2-27: Tests–Test Ea and guidance: Shock*. [https://webstore.iec.ch/en/publication/514]
- **[B5]** IEC 61000-4-2:2025, *Electromagnetic compatibility (EMC)–Part 4-2: Testing and measurement techniques–Electrostatic discharge immunity test*. [https://webstore.iec.ch/en/publication/68954]
- **[B6]** IEC 62368-1:2023, *Audio/video, information and communication technology equipment–Part 1: Safety requirements*. [https://webstore.iec.ch/en/publication/69308]
- **[B7]** IEEE Std C37.238-2017, *IEEE Standard Profile for Use of IEEE 1588 Precision Time Protocol in Power System Applications*, including IEEE Std C37.238a-2023. [https://standards.ieee.org/ieee/C37.238/6650/]
- **[B8]** ISO/IEC 17025:2017, *General requirements for the competence of testing and calibration laboratories*.

- **[B9]** ITU-T Recommendation G.8275.1/Y.1369.1 (02/2026), *Precision time protocol telecom profile for phase/time synchronization with full timing support from the network.* [https://www.itu.int/rec/T-REC-G.8275.1-202602-P/en]
- **[B10]** ITU-T Recommendation G.8275.2/Y.1369.2 (02/2026), *Precision time protocol telecom profile for phase/time synchronization with partial timing support from the network.* [https://www.itu.int/rec/T-REC-G.8275.2-202602-P/en]
- **[B11]** ITU-T Recommendation G.8271/Y.1366 (03/2020), *Time and phase synchronization aspects of telecommunication networks*, including Amendment 1 (08/2024). [https://www.itu.int/rec/T-REC-G.8271/en]

C.2 Metrology and Measurement References

- **[B12]** Joint Committee for Guides in Metrology, JCGM 100:2008, *Evaluation of measurement data–Guide to the expression of uncertainty in measurement.* [https://doi.org/10.59161/JCGM100-2008E]
- **[B13]** Joint Committee for Guides in Metrology, JCGM 200:2012, *International Vocabulary of Metrology–Basic and general concepts and associated terms (VIM)*, 3rd ed. [https://doi.org/10.59161/JCGM200-2012]
- **[B14]** Allan, D. W., "Time and Frequency (Time-Domain) Characterization, Estimation, and Prediction of Precision Clocks and Oscillators," *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control*, vol. 34, no. 6, pp. 647-654, Nov. 1987. [https://doi.org/10.1109/T-UFFC.1987.26997]
- **[B15]** Bregni, S., "Measurement of Maximum Time Interval Error for Telecommunications Clock Stability Characterization," *IEEE Transactions on Instrumentation and Measurement*, vol. 45, no. 5, pp. 900-906, Oct. 1996. [https://doi.org/10.1109/19.536708]
- **[B16]** Riley, W. J., and Howe, D. A., *Handbook of Frequency Stability Analysis*, NIST Special Publication 1065, 2008. [https://www.nist.gov/publications/handbook-frequency-stability-analysis]
- **[B17]** Novick, A. N., and Lombardi, M. A., "Practical Limitations of NTP Time Transfer," *Proceedings of the 2015 Joint Conference of the IEEE International Frequency Control Symposium and European Frequency and Time Forum*, 2015. [https://doi.org/10.1109/FCS.2015.7138909]

C.3 Open Implementations and Community Resources

- **[B18]** Open Hardware Repository, *White Rabbit project documentation.* [https://ohwr.org/projects/white-rabbit/]
- **[B19]** Network Time Foundation, *Network Time Protocol project documentation and reference implementation.* [https://www.ntp.org/]
- **[B20]** Open Compute Project Time Appliances Project, *TimeCard hardware, gateware, software, and documentation repository.* [https://github.com/Time-Appliances-Project/Time-Card]
- **[B21]** Open Compute Project, *Time Appliances Project public project page.* [https://www.opencompute.org/wiki/Time_Appliances_Project]

C.4 Distributed Systems and Transaction Processing References

- **[B22]** IBM, *ACID properties of transactions*, CICS Transaction Server for z/OS 5.4. [https://www.ibm.com/docs/en/cics-ts/5.4.0?topic=processing-acid-properties-transactions].
- **[B23]** Lamport, L., "Time, Clocks, and the Ordering of Events in a Distributed System," *Communications of the ACM*, vol. 21, no. 7, pp. 558-565, July 1978. [https://lamport.azurewebsites.net/pubs/time-clocks.pdf].
- **[B24]** Lamport, L., Shostak, R., and Pease, M., "The Byzantine Generals Problem," *ACM Transactions on Programming Languages and Systems*, vol. 4, no. 3, pp. 382-401, July 1982. [https://lamport.azurewebsites.net/pubs/byz.pdf].
- **[B25]** Gray, J., and Reuter, A., *Transaction Processing: Concepts and Techniques*, 1st ed., Morgan Kaufmann, 1992. [https://shop.elsevier.com/books/transaction-processing/gray/978-0-08-051955-5].
- **[B26]** van Steen, M., and Tanenbaum, A. S., *Distributed Systems*, 3rd ed., distributed-systems.net, 2017. [https://www.distributed-systems.net/index.php/books/ds3/].

C.5 Host Driver and Operating-System References

- [**B27**] Microsoft, *Supporting PnP and Power Management in Function Drivers*, Windows Driver Framework documentation. [https://learn.microsoft.com/en-us/windows-hardware/drivers/wdf/supporting-pnp-and-power-management-in-function-drivers].
- [**B28**] Microsoft, *How the Windows Time Service Works*, Windows Server documentation. [https://learn.microsoft.com/en-us/windows-server/networking/windows-time-service/how-the-windows-time-service-works].
- [**B29**] Apple Inc., *DriverKit*, Apple Developer Documentation. [https://developer.apple.com/documentation/driverkit].
- [**B30**] Apple Inc., *PCIDriverKit*, Apple Developer Documentation. [https://developer.apple.com/documentation/pcidriverkit].
- [**B31**] Linux kernel documentation, *PTP hardware clock infrastructure for Linux*. [https://docs.kernel.org/driver-api/ptp.html].

Annex D: Conformance Statement Proforma (Informative)

D.1 Purpose

This annex provides a common structure for the supplier conformance statement required by 4.7. It is a reporting aid and does not add, remove, or modify any conformance requirement.

The completed proforma should identify each claimed profile and optional feature, the implementation revision evaluated, and the evidence supporting each claim. A supplier may extend the tables when additional interfaces or application-defined profiles are declared.

D.2 Implementation identification

Field	Supplier response
Product or implementation name	To be completed
Supplier	To be completed
Hardware or implementation revision	To be completed
Firmware, gateware, and software revisions	To be completed
Host operating systems and versions	To be completed / not applicable
Driver, service, and time-provider revisions	To be completed / not applicable
Host API or ABI name and revision	To be completed / not applicable
Configuration identifier	To be completed
Form factor	To be completed
Date of statement	To be completed
Statement revision	To be completed
Supporting documentation	To be completed

D.3 Conformance profile claims

Profile and clause	Supplier claim and evidence
Base TimeCard (4.3)	Claimed: ***; evidence or limitation:***
Physical Timing Output (7.3.2)	Claimed: ***; evidence or limitation:***
PCIe Host Mapping (8.9)	Claimed: ***; evidence or limitation:***
Managed TimeCard (8.11.2)	Claimed: ***; evidence or limitation:***
Secure Infrastructure TimeCard (8.11.3)	Claimed: ***; evidence or limitation:***

D.4 Interface and optional-feature claims

Complete one record for each claimed interface or optional feature:

Field	Supplier response
Interface or feature	To be completed
Direction	Receive / providing / control / host
Standard, profile, or mapping revision	To be completed
Connector or logical endpoint	To be completed
Measurement point	To be completed
Instance identifier and stability scope	To be completed / not applicable
Discovery descriptor or baseline locator	To be completed / not applicable
Evidence or limitation	To be completed

Examples include GNSS, PTP, NTP, 1PPS, frequency outputs, Time of Day, IRIG, PCIe PTM, host time-stamping, reference selection, ensemble processing, firmware update, and sanitization.

D.5 Host-interface and driver declarations

Complete this section for each claimed host interface mapping:

Field	Supplier response
Host interface and mapping revision	To be completed
Supported operating systems, releases, and architectures	To be completed
Driver model, package, and signing or authorization status	To be completed

Field	Supplier response
Driver, service, host time-provider, and API or ABI revisions	To be completed
P3335 discovery-descriptor locator, revision, and encoding	To be completed
Supported PCI or implementation identities and board profiles	To be completed
`TC_INSTANCE_ID` derivation and stability scope	To be completed
`TC_SERIAL` namespace and availability	To be completed / not implemented
Host clock identifier, epoch, timescale, units, and adjustment behavior	To be completed
Host-time correlation method, measurement point, and maximum window	To be completed / not implemented
Sample-age, uncertainty or dispersion, and discipline-eligibility policy	To be completed / not implemented
Time-control ownership, arbitration, timeout, and recovery	To be completed / not implemented
Sleep, hibernation, wake, removal, reconnection, reboot, and driver-restart behavior	To be completed
Read, time-control, configuration, update, and security privileges	To be completed
Physical validation coverage and known limitations	To be completed

01 ## D.6 Performance declarations

Metric	Declaration record
Time accuracy	State: ***; point:*** ; bound: ***; conditions:*** ; method and uncertainty: ***; evidence:***
MTIE	State: ***; point:*** ; bound: ***; intervals:*** ; conditions: ***; method and uncertainty:*** ; evidence: _____

Metric	Declaration record
TDEV	State: ***; point:*** ; result or limit: ***; intervals:*** ; conditions: ***; method and uncertainty:*** ; evidence: _____
Frequency accuracy	State: ***; point:*** ; bound: ***; conditions:*** ; method and uncertainty: ***; evidence:***
ADEV	State: ***; point:*** ; result or limit: ***; intervals:*** ; conditions: ***; method and uncertainty:*** ; evidence: _____
Phase noise	State: ***; point:*** ; mask or points: ***; offset range:*** ; method and uncertainty: ***; evidence:***
Pulse timing variation	State: ***; point:*** ; statistic and bound: ***; bandwidth and sample count:*** ; uncertainty: ***; evidence:***
Holdover error	Entry condition: ***; point:*** ; bound versus elapsed time: ***; environment:*** ; method and uncertainty: ***; evidence:***
Transition behavior	Transition: ***; point:*** ; phase/frequency bound: ***; conditions:*** ; method and uncertainty: ***; evidence:***

01 ## D.7 Environment and lifecycle declarations

Category	Declaration and evidence
Operating and full-performance temperature	Range and location: ***; evidence:***
Storage and survival temperature	Range and condition: ***; evidence:***
Humidity and condensation	Range and condition: ***; evidence:***
Altitude or pressure	Range and derating: ***; evidence:***
Airflow or cooling assumptions	Requirement and condition: ***; evidence:***
Input power, sequencing, and interruption	Limits and behavior: ***; evidence:***
Host power-state and disconnect behavior	State retention and transition behavior: ***; evidence:***
Shock and vibration, if applicable	Profile and limit: ***; evidence:***
EMC and ESD conformity claims	Standard, edition, level, and configuration: ***; evidence:***
Reliability model or field-data basis	Metric and method: ***; evidence:***
Service-life items	Item, interval, and maintenance action: ***; evidence:***
Calibration interval or method	Interval, reference, method, and uncertainty: ***; evidence:***

02 ## D.8 Test evidence and deviations

03 Complete one record for each applicable clause, profile, or declared limit:

Field	Supplier response
Applicable clause or claim	To be completed
Verification method	Test / inspection / analysis / documentation
Report identifier	To be completed
Result	Pass / fail / not applicable
Deviation, waiver, or limitation	To be completed

Any deviation recorded in this table should state whether the affected feature remains within the conformance claim and identify the authority or rationale used for that determination.

Annex E: Host Operating-System Integration (Informative)

E.1 Purpose

This annex illustrates how the host-interface requirements in Clauses 5, 6, 8, and 9 can be applied on Windows, macOS, and Linux. It does not define an operating-system API, driver package, entitlement, device path, or command number, and it does not add to the conformance requirements.

The portable boundary is the P3335 discovery descriptor, baseline information model, mapping version, instance identity, host-time correlation record, lifecycle behavior, and time-control ownership. Operating-system mechanisms bind to that boundary. The Open Compute Project Time Appliances Project driver implementations provide useful implementation experience [B20].

E.2 Portable driver architecture

A portable host stack can be divided into four layers:

1. The operating system enumerates the PCIe function and assigns resources.
2. A privileged driver validates the P3335 discovery descriptor, bounds-checks resources, and owns hardware access.
3. A versioned host API exposes P3335 objects, events, and bounded time operations without exposing arbitrary MMIO to applications.
4. Optional services perform oscillator discipline, host-clock integration, monitoring, or firmware maintenance under explicit authorization and time-control ownership.

This separation keeps board profiles, register layouts, interrupt handling, and power transitions inside the component that can validate them. Applications consume stable semantic objects and can remain portable even when the operating-system transport differs.

E.2.1 Discovery and identity

The driver should obtain system-assigned PCI resources before reading the descriptor. It should validate the descriptor signature, version, lengths, generation, BAR bounds, resource spans, and capability indications before mapping or accessing an optional block.

One physical card can appear under a different enumeration index after reboot, removal, or inventory change. Configuration, calibration, and service selection should use `TC_INSTANCE_ID`, or `TC_SERIAL` when a persistent serial number is available. A device path can be retained as a secondary mapping-scoped selector, but its stability scope should be stated.

E.2.2 API and ABI boundary

A binary host API should use transport-enforced exact lengths or explicit structure lengths and versions. New fields are most safely appended, reserved fields remain zero, and unknown optional trailing fields are ignored only when the mapping declares that behavior. Read responses containing optional telemetry should preserve field-level validity instead of turning an absent field into a plausible zero.

Read-only monitoring and timing-affecting operations should be separate in both API access rights and user experience. Raw register access is useful for controlled engineering diagnostics but should not be the interoperability interface or an ordinary application privilege.

E.2.3 Time and discipline services

A host-time correlation should retain the two host bounds rather than only a derived midpoint. The correlation window, sample age, clock identity, timescales, card state, source validity, discontinuity generation, and uncertainty or dispersion policy are needed to decide whether the sample is suitable for host-clock discipline.

A background discipline service, command-line utility, monitoring application, and host time provider can otherwise compete for the same hardware. One component should own timing-affecting operations at a time, while read-only clients continue to observe state. Loss of the owner should lead to a bounded and observable ownership transition.

E.3 Windows considerations

Windows PCIe integration can use a Windows Driver Framework function driver to receive system-assigned resources and participate in Plug and Play and power transitions [B27]. The working-state entry, working-state exit, hardware-release, orderly-removal, and surprise-removal paths are natural points to start or stop interrupts, queues, polling, and MMIO access.

A versioned device interface or I/O control ABI can map driver operations to the P3335 objects. Read and write access masks can separate monitoring from time setting, discipline, configuration, and firmware update. The driver can expose one interface per TimeCard instance while retaining `TC_INSTANCE_ID` as the durable selection key.

Windows host-clock representations can differ from the card epoch and timescale. The driver or service should either normalize both timestamps into the P3335 `timestamp` representation or declare the host representation and conversion without losing the original correlation bounds.

Windows Time Service supports pluggable time providers that can obtain samples from hardware or software sources [B28]. A TimeCard provider should submit only samples that are discipline-eligible under 8.10.4. A Windows system-time jump, service restart, or card removal should invalidate older associations before another sample is offered.

Driver signing, package installation, service accounts, protected local interprocess communication, and production-release policy remain platform and deployment concerns. They should be recorded in the conformance statement when they affect availability, authorization, or the evaluated configuration.

E.4 macOS considerations

DriverKit provides user-space driver infrastructure, and PCIDriverKit provides access to custom PCI and PCIe hardware [B29], [B30]. A macOS implementation can package a DriverKit system extension with a host application and expose a versioned user-client API to authorized applications.

The DriverKit extension should validate the PCI identity, system-assigned BAR length, P3335 descriptor, and resource ranges before MMIO access. The user-client method dispatch can enforce exact input and output sizes, while the shared ABI preserves the same P3335 object semantics used on other operating systems.

A read-only control application can enumerate multiple driver services, select by TC_INSTANCE_ID, and display only fields marked valid. Driver activation, user-client authorization, and application signing are separate from the TimeCard information model and are best documented as platform bindings rather than encoded into the normative object vocabulary.

When macOS realtime readings bracket a card read, the mapping should identify that host clock and preserve both readings. Raw card time should remain labeled with its actual or unknown timescale until a trusted card-to-host timescale relationship and leap correction are available.

Sleep, wake, extension replacement, host-app removal, Thunderbolt or other tunneled-PCIe disconnection, and reconnection should be included in physical validation. Reconnection should produce a new discovery validation and discontinuity generation before time-control operations resume.

E.5 Linux considerations

The Linux PTP clock infrastructure provides a standard PHC abstraction and user-space clock operations [B31]. A Linux driver can map P3335 clock access to a PHC while exposing control, telemetry, events, and the discovery descriptor through documented kernel interfaces.

A `/dev/ptpN` number is an enumeration result and should not replace `TC_INSTANCE_ID` or `TC_SERIAL` in persistent configuration. Sysfs names, device nodes, and PCI paths can be useful secondary selectors when their stability scope is understood.

Linux cross-timestamp facilities can bind to `HOST_TIME_CORRELATION` by preserving the host clock identifier, bounds or measured interval, card timestamp, and uncertainty information. User-space tools should verify timescale and state before using a PHC to discipline the system clock.

Runtime power management, driver unbind and rebind, PCIe error recovery, hot plug, and host reboot should follow the same lifecycle rules as other platforms. Re-probing a known device identity does not remove the need to revalidate the active image and resource descriptor.

E.6 Cross-platform mapping summary

P3335 concept	Windows example	macOS example	Linux example
PCIe discovery	Framework-assigned resources plus validated descriptor	PCIDriverKit provider plus validated descriptor	PCI core resources plus validated descriptor
Instance selection	Device interface plus `TC_INSTANCE_ID`	Driver service plus `TC_INSTANCE_ID`	PHC or PCI path plus `TC_INSTANCE_ID`
Binary interface	Versioned device interface or I/O control ABI	Versioned DriverKit user-client ABI	Kernel API, ioctl, netlink, sysfs, or documented combination
Card time	Driver clock operation	DriverKit user-client clock operation	PHC operation
Host-time correlation	Bracketed host clock and card reading	Bracketed host clock and card reading	PTP cross-timestamp facility or equivalent
Host discipline	Eligible sample supplied to a Windows time provider	Privileged service using eligible samples	`phc2sys`-style service or equivalent

P3335 concept	Windows example	macOS example	Linux example
Time-control ownership	Driver or service lease	User-client or service claim	Kernel or daemon arbitration
Lifecycle	Plug and Play and power callbacks	DriverKit and system-extension lifecycle	PCI probe, remove, power-management, and error-recovery paths

E.7 Integration review checklist

Before a platform binding is used for a performance or conformance claim, an integrator should confirm:

- The exact hardware, firmware or gateware, driver, service, API or ABI, and operating-system revisions.
- Descriptor validation and fail-closed handling of unknown or ambiguous optional resources.
- Stable per-card selection and separation of per-card calibration and configuration.
- Atomic TimeCard reads and bounded host-time correlations with explicit timescales.
- Rejection of stale, discontinuous, source-invalid, or excessive-window discipline samples.
- Single-owner behavior for time setting and discipline.
- Bounded cancellation and resource revalidation through supported lifecycle transitions.
- Least-privilege separation of monitoring, time control, configuration, update, and security administration.
- Physical-card validation coverage for every claimed board profile and host connection type.